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An analytical cyclo-steady-state model for inter-rifting deformation at magmatic rifts

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SUMMARY
Inter-rifting deformation (the strain that accumulates between discrete magmatic episodes) remains the least-constrained phase of the rifting cycle. Existing kinematic frameworks represent the rift as a continuously opening dislocation in an elastic half-space, an assumption incompatible with both the known rheological structure of magmatic rifts and far-field geodetic constraints. We develop a fully analytical model for the surface fields generated by a periodic sequence of instantaneous dike intrusions in an elastic plate overlying a Newtonian viscous half-space. The governing equation reduces, in the Fourier domain, to a fractional (half-Laplacian) diffusion equation parameterized by a single characteristic velocity $a = EH/[2\eta(1-\nu^2)]$, where E, H, and ν are the Young's modulus, thickness, and Poisson's ratio of the brittle layer, and η is the asthenospheric viscosity. In the limit of many repeated intrusions the system reaches a Cyclo-Steady State (CSS) with a closed-form expression through the Digamma and log-Gamma functions, conserving far-field plate divergence by construction. We extend the model to a Maxwell visco-elastic asthenosphere via the elastic–visco-elastic correspondence principle, recovering two physically distinct relaxation regimes separated by the Maxwell relaxation time τ_R: near-field elastic rebound for $t \lesssim \tau_R$ and far-field viscous diffusion for $t \gg \tau_R$. We fit the viscous and Maxwell CSS directly to two magmatic rift systems, deriving τ_R from the same elastic modulus shared by plate and half-space rather than treating it as an independent free parameter, so the two rheologies are compared at equal parameter count. At Krafla (Foulger et al., 1992), the viscous CSS yields $\eta \approx 4 \times 10^{18}$ Pa s ($H = 10$ km, $E = 40$ GPa), with Maxwell rheology only marginally disfavoured ($\Delta\text{AIC} \approx 1$); at Asal (Vigny et al. 2007), the viscous CSS gives $\eta \approx 2 \times 10^{19}$ Pa s, a factor of five higher than at Krafla, and is decisively favoured over its Maxwell counterpart ($\Delta\text{AIC} \approx 30$) because the 25-yr record there samples enough of the post-diking transient to rule out a spurious elastic contribution. The viscosity contrast between the two sites reflects the anomalous sub-lithospheric heat flux beneath Iceland. The CSS framework provides a mechanically self-consistent description of inter-rifting deformation at any magmatic rift system, yielding viscosity estimates from a single geodetic velocity profile without auxiliary rheological assumptions.

Key words: continental rifting; inter-rifting deformation; fractional diffusion; visco-elastic relaxation; GPS geodesy; Iceland

1 INTRODUCTION

Continental rifting drives the transition from stable cratonic lithosphere to active oceanic spreading through rifting cycles that span a broad range of spatial and temporal scales (Corti 2009; Ebinger et al. 2013). These cycles are conventionally partitioned into three phases: elastic strain release through dike emplacement and normal faulting (Grandin et al. 2011; Maccaferri et al. 2016); post-diking visco-elastic relaxation, which decays on timescales of years to decades (Foulger et al. 1992; Nooner et al. 2009); and an inter-rifting phase during which the plates diverge at a secular rate set by large-scale mantle dynamics (Saria et al. 2014; Stamps et al. 2021). Geodetic campaigns in Iceland (Heki et al. 1993; Hofton et al. 1995) and the Afar (Vigny et al. 2007; Nooner et al. 2009; Grandin et al. 2011) have documented the first two phases in considerable detail. The mechanisms governing the build-up of extensional strain between discrete events, however, remain the least-constrained aspect of the rifting cycle.

The interseismic strain accumulation in rifts is poorly

constrained (Ebinger et al. 2013). Modelling the inter-rifting phase is conventionally done by transplanting kinematic frameworks developed for inter-seismic earthquake cycle. The standard approach to this problem borrows the backslip model developed for the interseismic earthquake cycle (Savage & Prescott 1978). The rift is assumed locked to depth H, below which it slips continuously, generating the basal shear forces needed to reproduce observed velocity profiles. Alternative formulations represent the rift as a dislocation in an elastic half-space that opens at a depth-dependent rate (LaFemina et al. 2005; Konca et al. 2018), or prescribe a basal shear traction to mimic far-field plate forcing (Ali et al. 2014). These models share a fundamental physical inconsistency: they ignore the rheological transition at the brittle–ductile boundary of magmatic rifts, where high heat flow and partial melt weaken the lower crust and uppermost mantle (Prodehl et al. 1997; Plasman et al. 2017). Geodetic and seismic constraints place this transition at 5–15 km depth in Iceland and the Afar (Stevens et al. 2021; Weinstein et al. 2017), implying visco-elastic relaxation times incompatible with the purely elastic response assumed by backslip models. A continuously opening dislocation is equally problematic: dike intrusions are finite, discrete events, and the geodetic record shows that each episode is followed by a localized transient decaying within a few years (Foulger et al. 1992; Reiss et al. 2022).

Efforts to model the post-diking transient have confronted the same boundary-condition obstacle from two directions. Analytical approaches, exemplified by the diffusion model of Foulger et al. (1992), reproduce the broad deformation pattern following individual intrusions but require an a priori specification of the viscous layer thickness as a free parameter (Figure 1). The same profile captures the transition from post-rifting relaxation to steady inter-rifting divergence directly. Twelve to seventeen years after the onset of rifting, the displacement field already traces a broad, arctangent-like transition rather than a transient confined to the rift. Both ends of the traverse settle onto distinct, non-zero rates instead of returning to zero far from the rift axis. Numerical visco-elastic simulations (Hofton et al. 1995; Hofton & Foulger 1996; Nooner et al. 2009) face an equivalent difficulty: in any finite domain, velocities must vanish at the lateral boundaries, a condition irreconcilable with the steady plate divergence observed in global reference frames (Fernandes et al. 2004; Altamimi et al. 2016). The result is a kinematic gap between the transient post-diking signal and the secular strain rates of global plate motion models (Kreemer et al. 2014; Stamps et al. 2018).

The theoretical foundations laid for cyclic strike-slip fault deformation (Nur & Mavko 1974; Savage & Prescott 1978; Spence & Turcotte 1979; Cohen & Kramer 1984) suggest a different perspective. Inter-rifting deformation is not a distinct physical process but the emergent superposition of visco-elastic responses to an infinite history of discrete dike intrusions. The apparent steadiness of inter-rifting velocities arises naturally from the time-averaging of decaying transients, with no need for a continuously creeping source.

We develop an analytical model for an elastic crust overlying a viscous half-space (Hetland 2006) in which dike intrusions recur periodically at interval T and each open the rift by $2b$, where b is the dike half-opening. The governing integro-differential equation reduces, after a Fourier transform, to a fractional diffusion equation (Metzler & Klafter 2000; Samko & Ross 1993) parameterized by a single characteristic velocity $a = EH/[2\eta(1-\nu^2)]$. In the limit of many repeated intrusions, the system reaches a Cyclo-Steady State (CSS) expressed exactly through the Digamma and log-Gamma functions, which satisfies the far-field plate divergence constraint by construction. We then extend the model to a Maxwell visco-elastic asthenosphere via the elastic–visco-elastic correspondence principle (Hetland & Hager 2005; Barbot & Fialko 2010), recovering two physically distinct relaxation regimes separated by the Maxwell timescale $\tau_R = \eta/\mu_m$. We apply both models to two geodetic datasets: horizontal displacement profiles from the Krafla rift, Iceland, recorded 8–11 years after the 1975–1984 rifting sequence (Foulger et al. 1992), and a 25-year baseline time series from the Asal rift, Djibouti (Vigny et al. 2007). Together, they constrain asthenospheric viscosity across settings separated by a factor of five, and show that decadal post-rifting transient and secular plate divergence are the short- and long-term expression of a singly underlying process. and secular plate divergence.

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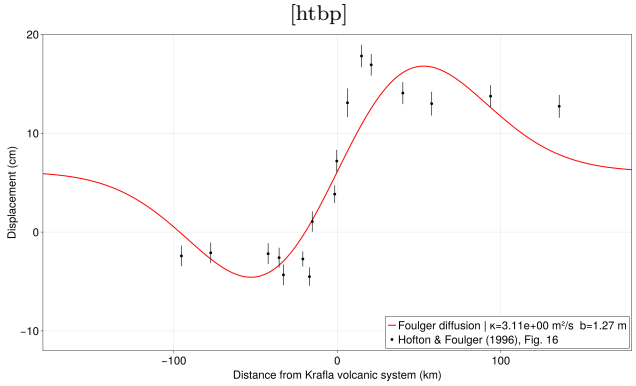


Figure 1. Figure recreated from Hofton & Foulger (1996), their Figure 16: rift-perpendicular horizontal displacements at Krafla, 1987–1992, with 1σ error bars (black circles), and the Foulger et al. (1992) diffusion model $u(x,\tau) = b\,\mathrm{erfc}[x/(2\sqrt{\kappa\tau})]$, differenced between $\tau_i = 12$ and $\tau_f = 17$ yr after the onset of the 1975–1984 rifting sequence, at the $\kappa = 3.11$ m^2 s^{-1}, $b = 1.27$ m best-fit parameters reported by Hofton & Foulger (red curve; a constant reference-frame offset, not a free shape parameter of the model, is added to match the digitized profile). The systematic misfit at near- and far-field stations simultaneously ($\chi^2_r \approx 10$) leads Hofton & Foulger (1996) to argue that a viscoelastic half-space, rather than simple diffusion, is needed to explain the post-rifting deformation field.

2 MODEL

2.1 Physical setup and governing equations

The model represents the lithosphere as a coupled mechanical system: a thin elastic plate of thickness H (the seismogenic crust) overlying a Newtonian viscous half-space of viscosity η representing the asthenosphere (e.g., Hetland & Hager 2005; Hetland 2006). The horizontal coordinate x is perpendicular to the rift axis; the vertical coordinate z is oriented downward, with the free surface at $z = 0$ and the crust–mantle interface at $z = H$. Under plane-strain con-

ditions with negligible vertical stress ($\sigma_{zz} \approx 0$), the depth-averaged horizontal stress in the elastic layer is related to the horizontal surface displacement $u(x,t)$ through Hooke's law:

$$\bar{\sigma}_{xx} = \frac{E}{1-\nu^2}\frac{\partial u}{\partial x}, \tag{1}$$

where E is Young's modulus and ν is Poisson's ratio. The factor $(1-\nu^2)^{-1}$ accounts for the lateral confinement of plane strain. Mechanical equilibrium of the plate requires the horizontal gradient of the depth-averaged stress to be balanced by the shear traction exerted at its base by the underlying viscous mantle:

$$H\frac{\partial \bar{\sigma}_{xx}}{\partial x} + \sigma_{xz}^{\mathrm{BOT}} = 0. \tag{2}$$

The rheological character of the asthenosphere enters through the constitutive relation for the basal traction $\sigma_{xz}^{\mathrm{BOT}}$. For a Newtonian viscous half-space, the shear stress at x depends on the surface velocity at *all* other points through the non-local Stokes kernel (Savage & Prescott 1978):

$$\sigma_{xz}^{\mathrm{BOT}}(x,t) = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2}\frac{\partial u}{\partial t}(x',t)\,dx', \tag{3}$$

interpreted as Hadamard finite part. This non-local coupling reflects the long-range character of viscous flow in the half-space: a velocity perturbation at x' decays as $(x-x')^{-2}$ rather than being confined to a finite diffusion radius. In magmatically active rifts, where high heat flow and partial melt weaken the lower crust (Holtzman 2016; Plasman et al. 2017), this coupling operates across the entire plate boundary system.

Foulger et al. (1992) start from the same force balance but close it differently. Rather than coupling the elastic layer to an infinite viscous half-space through the non-local Stokes kernel (3), they place a viscous channel of finite thickness H_v over a rigid base. Its basal traction is then treated as local, simple-shear drag, $\sigma_{xz}^{\mathrm{BOT}} \sim (\eta/H_v)\,\partial u/\partial t$, in place of the convolution in (3). Substituted into (2), this closure collapses the governing equation to an ordinary, Fickian diffusion equation for the surface displacement, $\partial u/\partial t = \kappa\,\partial^2 u/\partial x^2$. The diffusivity $\kappa \propto EHH_v/\eta$ this introduces aggregates the elastic-layer properties with the assumed viscous-channel thickness H_v, rather than constraining a half-space viscosity directly. Reproducing a given deformation pattern therefore requires an a priori choice of H_v at each site – the limitation already noted in Section 1. Section 2.2 derives what the non-local closure (3) yields in its place: an infinite half-space, rather than a finite channel, replaces the ordinary Laplacian of Foulger et al.'s diffusion equation with the half-order fractional Laplacian $(-\Delta)^{1/2}$ of Equation 9. The two models therefore differ in the order of the governing diffusion equation itself, not only in its diffusivity.

Substituting (1) into (2) and using (3) yields the governing integro-differential equation for u:

$$\frac{EH}{1-\nu^2}\frac{\partial^2 u}{\partial x^2} + \int_{-\infty}^{+\infty}\frac{2\eta}{\pi(x-x')^2}\frac{\partial u}{\partial t}(x',t)\,dx' = 0. \tag{4}$$

Dike intrusions enter as initial conditions: each event opens the rift instantaneously by b, offsetting the block on one side of the axis relative to the other (Figure 2). Section 2.3 and Appendix A show that this offset subsequently redistributes

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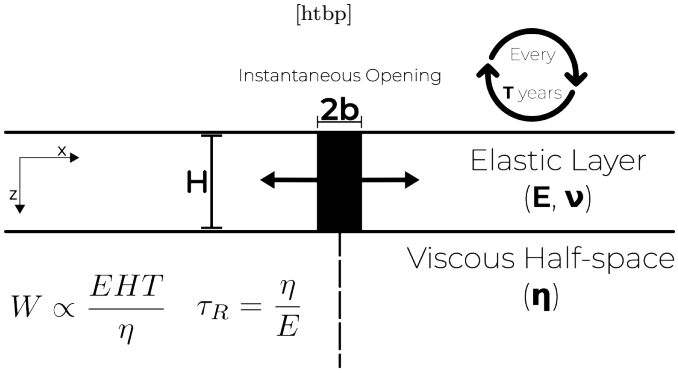


Figure 2. Model geometry. The brittle elastic plate (thickness H, Young's modulus E, Poisson's ratio ν) overlies a Newtonian viscous half-space (viscosity η). Dike intrusions open the rift by b at intervals T (Appendix A); the surface displacement $u(x,t)$ is positive away from the rift axis. The parameter $a = EH/[2\eta(1-\nu^2)]$ sets the characteristic velocity that couples elastic and viscous deformation.

into a symmetric $\pm b/2$ split about the pre-intrusion position far from the rift, while the opening measured at the intrusion site itself remains exactly b at all times.

2.2 Fourier reduction and the characteristic velocity

Equation (4) is non-local in x and not tractable in physical space. A spatial Fourier transform resolves this, because the Stokes convolution diagonalizes in wavenumber space. Defining the transform pair

$$\hat{u}(k,t) = \int_{-\infty}^{+\infty} u(x,t)e^{-ikx}\,dx, \qquad u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty}\hat{u}(k,t)e^{ikx}\,dk,$$

the transform of $\partial^2 u/\partial x^2$ is $-k^2\hat{u}$. The transform of the Stokes kernel convolution is the classical result (Savage & Prescott 1978):

$$\mathcal{F}\left\{\int_{-\infty}^{+\infty}\frac{2\eta}{\pi(x-x')^2}\frac{\partial u}{\partial t}\,dx'\right\} = -2\eta|k|\,\frac{\partial \hat{u}}{\partial t}. \tag{5}$$

Applying the Fourier transform to (4) and using (5):

$$-\frac{EH}{1-\nu^2}k^2\hat{u} - 2\eta|k|\frac{\partial \hat{u}}{\partial t} = 0. \tag{6}$$

Rearranging and introducing the characteristic velocity

$$\boxed{a := \frac{EH}{2\eta(1-\nu^2)}}, \tag{7}$$

equation (6) simplifies to a decoupled first-order ODE for each wavenumber k:

$$\frac{\partial \hat{u}}{\partial t} = -a\,|k|\,\hat{u}. \tag{8}$$

The parameter a encodes the full elastic–viscous coupling of the system: large a (stiff crust, low viscosity) implies rapid redistribution of post-diking strain over broad spatial scales; small a implies slow, locally confined relaxation. In physical space, the operator $|k|$ corresponds to the fractional half-Laplacian $(-\Delta)^{1/2}$, so (8) is equivalently written as the fractional diffusion equation (Metzler & Klafter 2000; Samko &

Ross 1993; Samko et al. 1993)

$$\frac{\partial u}{\partial t} = -a\,(-\Delta)^{1/2} u. \tag{9}$$

This order is not a modelling choice. The fractional Laplacian admits a representation as a Dirichlet-to-Neumann map for a harmonic extension problem into an auxiliary half-space, weighted by y^{1-2s}; the weight vanishes, and the extension becomes the ordinary Laplace equation, only at $s = 1/2$ (Caffarelli & Silvestre 2007; Daoud & Laamri 2022). That is the same half-space potential problem generating the Stokes kernel itself, so the half-order non-locality in (9) is forced by the geometry of the elastic-plate–viscous-half-space coupling, not fitted to the data. Solving the full two-dimensional Stokes flow beneath the plate, governed by the biharmonic rather than the harmonic equation, still recovers the same interface operator $-2\eta(-\Delta)^{1/2}$ exactly once the interface stress is extracted.

This contrasts with classical Fickian diffusion: the half-order operator, a direct consequence of the non-local Stokes kernel, produces Lorentzian rather than Gaussian spatial profiles, with tails that decay as $1/x$ rather than exponentially. The solution of (8) for arbitrary initial data is:

$$\hat{u}(k,t) = \hat{u}(k,0)\, e^{-a|k|t}. \tag{10}$$

2.3 Step response to a single dike intrusion

The simplest building block of the model is the deformation produced by a single instantaneous dike intrusion at $t = 0$, which opens the rift by b. Because the governing equation (4) constrains u only through its gradient, the reference level from which this offset is measured is arbitrary, and the initial condition can be written without loss of generality as a Heaviside displacement:

$$u(x,0^+) = b\,\mathcal{H}(-x) = \begin{cases} 0\,, & x > 0 \\ b\,, & x < 0 \end{cases}. \tag{11}$$

Applying the Fourier transform to (11) gives $\hat{u}(k,0) = b\int_{-\infty}^{0} e^{-ikx}\,dx$, which for real k is treated in the distributional sense by adding a convergence regulator $\varepsilon > 0$:

$$\int_{-\infty}^{0} e^{-ikx}\,dx = \lim_{\varepsilon\to 0}\int_{-\infty}^{0} e^{(-ik+\varepsilon)x}\,dx = \lim_{\varepsilon\to 0}\frac{1}{\varepsilon - ik}.$$

By the Sokhotski–Plemelj theorem (Ablowitz & Fokas 2003),

$$\lim_{\varepsilon\to 0^+}\frac{1}{x \pm i\varepsilon} = \mathcal{P}\left(\frac{1}{x}\right) \mp i\pi\delta(x), \tag{12}$$

where $\mathcal{P}$ denotes the Cauchy principal value. Multiplying numerator and denominator by i recasts the regulator as $+i\varepsilon$,

$$\frac{1}{\varepsilon - ik} = \frac{i}{i\varepsilon + k} = \frac{i}{k + i\varepsilon},$$

so that (12) applies directly with $x = k$ and the upper sign:

$$\begin{aligned}\lim_{\varepsilon\to 0^+}\frac{1}{\varepsilon - ik} &= \lim_{\varepsilon\to 0^+}\frac{i}{k + i\varepsilon} \\ &= i\left[\mathcal{P}\left(\frac{1}{k}\right) - i\pi\delta(k)\right] \\ &= \pi\delta(k) + \frac{i}{k},\end{aligned} \tag{13}$$

where the principal-value symbol on $1/k$ is dropped from here on, as is standard, since it is understood implicitly whenever $1/k$ appears next to a $\delta(k)$ term. Hence the initial Fourier spectrum is:

$$\hat{u}(k,0) = b\pi\delta(k) + \frac{ib}{k}. \tag{14}$$

Substituting (14) into (10) and noting that $\delta(k)$ is active only where $e^{-a|k|t} = 1$:

$$\hat{u}(k,t) = b\pi\delta(k) + \frac{ib}{k}\,e^{-a|k|t}. \tag{15}$$

The inverse Fourier transform splits into two contributions, $u(x,t) = I_1 + I_2$. The delta-function term gives a constant background $I_1 = b/2$; for I_2, the parity of $\sin(kx)/k$ yields:

$$I_2 = -\frac{b}{\pi}\int_0^\infty \frac{\sin(kx)}{k}\,e^{-akt}\,dk. \tag{16}$$

This is a standard Dirichlet-type integral. Applying the complementary arctangent identity $\arctan(z) + \arctan(1/z) = \pi/2$, the step displacement is:

$$\boxed{u^{\mathrm{STEP}}(x,t) = \frac{b}{\pi}\arctan\left(\frac{at}{x}\right).} \tag{17}$$

Differentiating with respect to t, the step velocity is a Lorentzian profile that broadens in time:

$$v^{\mathrm{STEP}}(x,t) = \frac{ba}{\pi}\,\frac{x}{x^2 + a^2t^2}. \tag{18}$$

Both expressions satisfy the correct limits: $u^{\mathrm{STEP}} \to b/2$ and $v^{\mathrm{STEP}} \to 0$ as $t \to \infty$, consistent with far-field plate divergence. This limit reflects the profile's self-similarity: u^{STEP} depends on x and t only through the ratio $x/(at)$. At any fixed x, that ratio vanishes as $t \to \infty$, so the profile relaxes to the midpoint of its own jump, $b/2$, rather than to b. This is not a loss of the imposed opening: Appendix A shows that the discontinuity at the intrusion site itself remains exactly b for all $t > 0$, so the amplitude b fitted in §3.2 and §3.3 is directly the physical dike-opening width, not half of it. The velocity peak migrates outward as at, with amplitude decaying as $1/t$. On the connection between (17) and the Foulger et al. (1992) diffusion model, see the fractional-Laplacian annex, §3.2.

2.4 Cyclic accumulation and the Cyclo-Steady State

2.4.1 Finite sum over past intrusions

A rift that has experienced $n = \lfloor t/T \rfloor$ dike intrusions at regular intervals of T years accumulates displacement by superposition of individual step responses:

$$u(x,t) = \frac{b}{\pi}\sum_{j=1}^{n}\arctan\left(\frac{a(t - jT)}{x}\right). \tag{19}$$

Differentiating with respect to t, the corresponding velocity field is:

$$v(x,t) = \frac{ba}{\pi}\sum_{j=1}^{n}\frac{x}{x^2 + a^2(t - jT)^2}. \tag{20}$$

2.4.2 The CSS limit and Digamma representation

After many cycles the system enters a periodic state, the rift analog of the cyclic steady state described for the interseismic earthquake cycle (Thatcher & Rundle 1984; Hetland & Hager 2006). Introducing $\tau \in [0, T)$ as the time elapsed since the most recent intrusion, the Cyclo-Steady State (CSS) velocity is obtained by extending (20) to an infinite history:

$$v_{\mathrm{CSS}}(x,\tau) = \frac{ba}{\pi} \sum_{j=1}^{\infty} \frac{x}{x^2 + a^2(\tau - jT)^2}. \tag{21}$$

Because this series converges absolutely, it admits a closed-form expression via the Digamma function $\psi(z)$. Using the partial-fraction identity

$$\frac{x}{x^2 + a^2(\tau - jT)^2} = \frac{1}{2i}\left[\frac{1}{a(\tau - jT) - ix} - \frac{1}{a(\tau - jT) + ix}\right],$$

factoring $-aT$ from each denominator and substituting into (21), the double sum takes the form of differences of Digamma series. Using the representation $\psi(z+1) = -\gamma + \sum_{n=1}^{\infty}(1/n - 1/(n+z))$, the Euler–Mascheroni constant γ and the $1/n$ terms cancel when the two conjugate series are differenced. With $Z = \tau/T + ix/(aT)$ and the identity $\psi(Z) - \psi(\bar{Z}) = 2i\,\mathrm{Im}\{\psi(Z)\}$, the CSS velocity collapses to:

$$\boxed{v_{\mathrm{CSS}}(x,\tau) = \frac{2b}{\pi T}\,\mathrm{Im}\left\{\psi\left(\frac{\tau}{T} + i\frac{x}{aT}\right)\right\}.} \tag{22}$$

2.4.3 CSS displacement and far-field conservation

Integrating (22) over τ, and using the antiderivative $d[\ln\Gamma(z)]/dz = \psi(z)$, the CSS displacement is:

$$\boxed{u_{\mathrm{CSS}}(x,\tau) = \frac{2b}{\pi}\,\mathrm{Im}\left\{\ln\Gamma\left(\frac{\tau}{T} + i\frac{x}{aT}\right) - \ln\Gamma\left(i\frac{x}{aT}\right)\right\}.} \tag{23}$$

Over one complete cycle, the cumulative displacement at any fixed $x > 0$ equals the dike half-opening:

$$\int_0^T v_{\mathrm{CSS}}(x,\tau)\,d\tau = b, \qquad \forall\, x > 0. \tag{24}$$

This is not imposed as a boundary condition but follows analytically from the Digamma representation: $\Gamma(1+iq) = iq\,\Gamma(iq)$ implies $\mathrm{Im}\{\ln\Gamma(1+iq) - \ln\Gamma(iq)\} = \pi/2$, giving the integral b. The CSS thus reconciles high-frequency magmatic pulses with secular divergence without unphysical boundary conditions, satisfying the far-field constraint of global plate motion models (Saria et al. 2014; Stamps et al. 2021) by construction.

The spatial width of the CSS deformation zone scales as $aT \propto EHT/\eta$. Because the CSS depends on the rift parameters (b, T) only through the product aT, geodetic profiles of inter-rifting deformation directly constrain the ratio EH/η, providing a new route to lithospheric characterization in systems where T can be independently estimated (Figure 3a). The finite-sum approximation to the CSS converges to within the measurement noise of a typical geodetic network after approximately 10 rifting cycles (Figure 3b); for systems with fewer events on record, the CSS should be replaced by the explicit sum in (19), and only the step-response regime of Section 2.3 applies.

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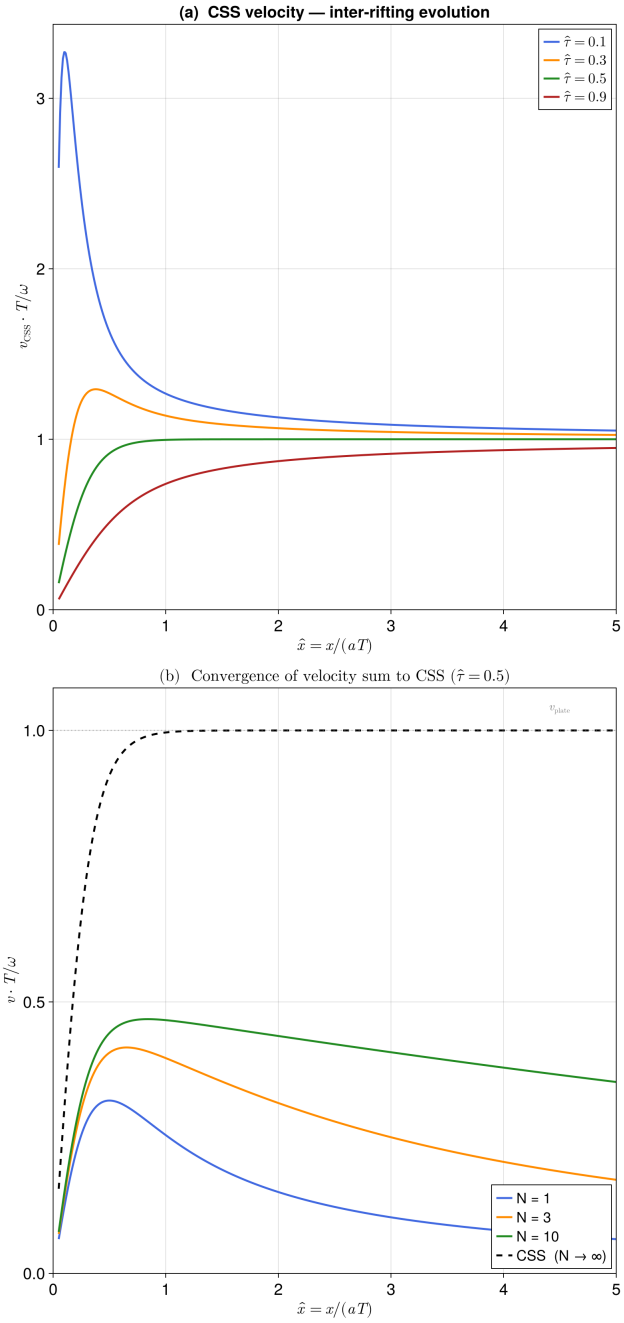


Figure 3. Dimensionless CSS velocity and convergence from finite rifting histories. All quantities are normalised by b and T; distances by aT. **(a)** CSS velocity $v_{\mathrm{CSS}} \cdot T/b$ versus $\hat{x} = x/(aT)$ at four inter-rifting phases $\hat{\tau} = \tau/T$. The velocity peak width is locked at aT regardless of phase; all profiles converge to $v_{\mathrm{plate}} = b/T$ (dotted line) in the far field. **(b)** Velocity at $\hat{\tau} = 0.5$ for $N = 1$, 3, and 10 dike events (solid curves) compared with the CSS limit (dashed). The $N = 1$ profile does not recover the far-field plate velocity; convergence to the CSS requires $\sim$10 cycles.

2.5 Extension to a Maxwell visco-elastic asthenosphere

2.5.1 Constitutive model and the correspondence principle

The Maxwell relaxation time $\tau_R := \eta/\mu_m$ (where μ_m is the mantle shear modulus) separates two regimes: on timescales $t \ll \tau_R$ the asthenosphere responds elastically; on timescales $t \gg \tau_R$ it flows viscously (Bürgmann & Dresen 2008). We model this by replacing the Newtonian half-space with a Maxwell body: a Hookean spring and a Newtonian dashpot in series. The 1D constitutive relation for deviatoric stress τ_{ij} and strain ε is:

$$\dot{\varepsilon} = \frac{\dot{\tau}}{2\mu_m} + \frac{\tau}{2\eta}. \tag{25}$$

To adapt the Stokes kernel (3) we employ the elastic–visco-elastic correspondence principle (Hetland & Hager 2005; Barbot & Fialko 2010): in the Laplace domain, static moduli are replaced by their complex, frequency-dependent counterparts. Applying the Laplace transform $\tilde{f}(s) = \mathcal{L}\{f(t)\}$ to (25) with zero initial conditions:

$$s\tilde{\varepsilon} = \tilde{\tau}\left(\frac{s}{2\mu_m} + \frac{1}{2\eta}\right). \tag{26}$$

Solving for stress gives $\tilde{\tau} = 2\tilde{\mu}(s)\tilde{\varepsilon}$, where the effective frequency-dependent shear modulus of the Maxwell body is:

$$\tilde{\mu}(s) = \frac{\mu_m \eta\, s}{\mu_m + \eta s} = \mu_m \frac{\tau_R s}{1 + \tau_R s}. \tag{27}$$

2.5.2 The visco-elastic Stokes kernel

For a purely elastic half-space the basal traction relates to surface displacement through a kernel proportional to μ_m:

$$\sigma_{xz}^{\mathrm{BOT,\ el}}(x) = \int_{-\infty}^{+\infty} \frac{2\mu_m}{\pi(x-x')^2}\, u(x')\, dx'. \tag{28}$$

Applying the correspondence principle (replacing μ_m with $\tilde{\mu}(s)$ in the Laplace domain) and factoring out $\eta = \mu_m \tau_R$:

$$(1 + \tau_R s)\, \tilde{\sigma}_{xz}^{\mathrm{BOT}}(x,s) = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2} \left[s\tilde{u}(x',s)\right] dx'. \tag{29}$$

Inverting to the time domain (multiplication by s in the Laplace domain corresponds to $\partial/\partial t$) yields the visco-elastic Stokes kernel:

$$\sigma_{xz}^{\mathrm{BOT}} + \tau_R \frac{\partial \sigma_{xz}^{\mathrm{BOT}}}{\partial t} = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2} \frac{\partial u(x',t)}{\partial t}\, dx'. \tag{30}$$

In the limit $\tau_R \to 0$, (30) reduces to the purely viscous Stokes kernel (3).

2.5.3 Governing equation in Fourier space

The equilibrium equation (2) expresses the basal traction in terms of the elastic crust displacement:

$$\sigma_{xz}^{\mathrm{BOT}} = -\frac{EH}{1-\nu^2} \frac{\partial^2 u}{\partial x^2}. \tag{31}$$

Substituting (31) and its time derivative into (30), then applying the spatial Fourier transform using (5):

$$\frac{EH}{1-\nu^2} k^2 \hat{u} + \tau_R \frac{EH}{1-\nu^2} k^2 \frac{\partial \hat{u}}{\partial t} = -2\eta|k| \frac{\partial \hat{u}}{\partial t}. \tag{32}$$

Dividing by 2η, using (7), and collecting $\partial\hat{u}/\partial t$ terms gives the visco-elastic analogue of (8):

$$\frac{\partial \hat{u}}{\partial t} = -\left(\frac{a|k|}{1 + \tau_R a|k|}\right) \hat{u}. \tag{33}$$

The denominator $1+\tau_R a|k|$ acts as a low-pass filter on the relaxation rate, slowing the decay of high-wavenumber (near-field) features relative to the purely viscous case.

2.5.4 Step response and asymptotic behavior

Equation (33) is separable in k, yielding a decaying exponential for each wavenumber:

$$\hat{u}(k,t) = \hat{u}(k,0) \exp\left(-\frac{a|k|}{1 + \tau_R a|k|}\, t\right). \tag{34}$$

Applying the Heaviside initial condition (14):

$$\hat{u}(k,t) = b\pi\delta(k) + \frac{ib}{k} \exp\left(-\frac{a|k|}{1 + \tau_R a|k|}\, t\right). \tag{35}$$

Taking the inverse Fourier transform and following the same parity argument as in Section 2.3, the Maxwell step displacement requires numerical integration:

$$u^{\mathrm{MAX}}(x,t) = \frac{b}{2} - \frac{b}{\pi} \int_0^{+\infty} \frac{\sin(kx)}{k} \exp\left(-\frac{akt}{1 + \tau_R ak}\right) dk, \tag{36}$$

since the rational exponent prevents the closed-form Dirichlet evaluation of Section 2.3. The surface velocity follows by differentiating (36) with respect to t:

$$v^{\mathrm{MAX}}(x,t) = \frac{ba}{\pi} \int_0^{+\infty} \frac{\sin(kx)}{1 + \tau_R ak} \exp\left(-\frac{akt}{1 + \tau_R ak}\right) dk. \tag{37}$$

Both integrals are evaluated numerically using adaptive Gauss-Kronrod quadrature (Laurie 1997), with integration breakpoints placed at the successive zeros of $\sin(kx)$ (i.e. $k = n\pi/x$, $n = 1, 2, \ldots$) to partition the oscillatory integrand into well-behaved sub-intervals. The upper truncation wavenumber is set adaptively to ensure the exponential factor is negligible at the integration cutoff.

The exponent in (36) reveals two physically distinct regimes. In the long-wavelength limit ($k \to 0$ or $t \to \infty$), $1 + \tau_R ak \approx 1$ and the exponential reduces to $\exp(-akt)$, recovering the purely viscous Dirichlet integral of Section 2.3. The elastic memory of the asthenosphere fades at large distances and late times, and the model converges to the Newtonian viscous limit. In the short-wavelength limit ($k \to \infty$ or $t \to 0$), $1 + \tau_R ak \approx \tau_R ak$ and the exponent saturates:

$$\lim_{k\to\infty} \left(-\frac{akt}{1 + \tau_R ak}\right) = -\frac{t}{\tau_R}. \tag{38}$$

At early times $t \ll \tau_R$, the exponential approaches unity and no near-field deformation occurs outside the rift interface. For $t \sim \tau_R$, near-field deformation decays exponentially at the fixed rate $1/\tau_R$, representing pure Maxwell relaxation of the mantle independent of the elastic crust. Together, (37) and (38) demonstrate that the Maxwell model introduces a short-time, near-field stiffening relative to the viscous model: both converge to the same Cyclo-Steady State at $t \gg \tau_R$ (Section 2.6).

2.6 Maxwell Cyclo-Steady State

2.6.1 Infinite sum and Fourier periodicity condition

The Maxwell CSS is the superposition of step-response velocities from all preceding intrusions, by the same argument as in Section 2.4:

$$v_{\mathrm{CSS}}^{\mathrm{MAX}}(x,\tau) = \sum_{j=0}^{\infty} v^{\mathrm{STEP,MAX}}(x,\, \tau + jT). \tag{39}$$

Writing $v^{\mathrm{STEP,MAX}}$ in Fourier space (Section 2.5) as $\frac{2ba}{\pi}\int_0^\infty \sin(kx)\, e^{-\alpha(k)(\tau+jT)}\, dk$, with the effective Maxwell decay rate

$$\alpha(k) := \frac{ak}{1+\tau_R ak}, \tag{40}$$

each past intrusion contributes a factor $e^{-\alpha(k)jT}$. The series converges for all $k > 0$ and the geometric sum $\sum_{j=0}^{\infty} e^{-\alpha(k)jT} = (1-e^{-\alpha(k)T})^{-1}$ can be performed inside the Fourier integral. The same result follows directly from the requirement that each Fourier mode satisfy the periodicity condition $\hat{u}(k,\tau+T) = \hat{u}(k,\tau) + ib/k$, which admits the unique solution

$$\hat{u}(k,\tau) = \frac{ib/k}{1-e^{-\alpha(k)T}}\, e^{-\alpha(k)\tau}, \qquad \tau \in [0,T). \tag{41}$$

2.6.2 Physical-space representation, conservation, and limits

Inverting (41) gives the Maxwell CSS velocity and displacement:

$$v_{\mathrm{CSS}}^{\mathrm{MAX}}(x,\tau) = \frac{2ba}{\pi}\int_0^\infty \frac{\sin(kx)}{1+\tau_R ak}\,\frac{e^{-\alpha(k)\tau}}{1-e^{-\alpha(k)T}}\, dk, \tag{42}$$

$$u_{\mathrm{CSS}}^{\mathrm{MAX}}(x,\tau) = \frac{2b}{\pi}\int_0^\infty \frac{\sin(kx)}{k}\,\frac{1-e^{-\alpha(k)\tau}}{1-e^{-\alpha(k)T}}\, dk. \tag{43}$$

Integrating the velocity over one full period, using $\int_0^T e^{-\alpha\tau}/(1-e^{-\alpha T})\, d\tau = 1/\alpha$ and the algebraic identity $a/[(1+\tau_R ak)\,\alpha(k)] = 1/k$, gives

$$\int_0^T v_{\mathrm{CSS}}^{\mathrm{MAX}}(x,\tau)\, d\tau = \frac{2b}{\pi}\int_0^\infty \frac{\sin(kx)}{k}\, dk = b, \tag{44}$$

where the last step is the Dirichlet integral. The full rift half-opening is recovered every cycle, independent of τ_R. In the limit $\tau_R \to 0$, $\alpha(k) \to ak$ and $(1+\tau_R ak)^{-1} \to 1$, and (42) reduces to the viscous CSS Fourier integral whose closed form is the Digamma series of Section 2.4. For $\tau_R > 0$, the saturating denominator $1+\tau_R ak$ prevents this reduction and (42) is evaluated numerically.

2.6.3 Constraining τ_R without an extra free parameter

Section 3 fits the Maxwell CSS to real data alongside its purely viscous counterpart. Leaving τ_R free alongside $(a,\, b)$ would let the extra parameter absorb noise rather than test whether Maxwell rheology genuinely improves the fit, since a model with an additional free parameter is guaranteed to reach at least as low a χ^2 regardless of whether that parameter is physically warranted. We instead derive τ_R from a itself, so that the Maxwell CSS carries exactly as many free parameters as the viscous CSS it is compared against.

The half-space's Maxwell shear modulus μ_m (Equation 27) has so far been left general. We adopt the simplest closure available without additional data: the half-space shares the same shear modulus as the elastic plate it underlies,

$$\mu_m = \mu_{\mathrm{plate}} := \frac{E}{2(1+\nu)}. \tag{45}$$

Combined with $\eta = EH/[2a(1-\nu^2)]$, the inversion of the characteristic velocity (7), this gives

$$\boxed{\tau_R = \frac{\eta}{\mu_m} = \frac{H}{a(1-\nu)}}, \tag{46}$$

a relaxation time set entirely by the already-fitted characteristic velocity a and the canonical geometric parameters H, ν (Section 4.2), with no independent estimate of μ_m required. This is a simplifying assumption, not a physical necessity – the mantle shear modulus need not equal the crust's – but it is the natural default in the absence of independent constraints, and it keeps the viscous-versus-Maxwell comparison in Section 3 an honest, equal-parameter test of functional form rather than one biased toward the model with more free parameters.

3 RESULTS: APPLICATIONS TO MAGMATIC RIFT SYSTEMS

Post-diking geodetic records from two contrasting rift environments test the models developed in Section 2 and constrain sub-lithospheric viscosity across distinct tectonic settings. We apply the viscous step response (Section 2.3) to a rift-perpendicular differenced displacement profile from the 1975–1984 Krafla rifting sequence in NE Iceland (Fouler et al. 1992) and to a 25-year baseline time series combined with a rift-perpendicular velocity profile from the 1978 Asal–Ghoubbet episode in the Afar Triangle (Vigny et al. 2007). Both datasets fall within the single-event step-response regime: at 8–11 yr post-rifting, Krafla has experienced less than one recurrence interval, and the Asal record ends before a subsequent major rifting episode. Together, the two sites span a factor-of-five contrast in best-fit characteristic velocity a and probe the rheological contrast between a plume-fed mid-ocean rift and an early-stage continental rift.

3.1 Dimensionless form and model behavior

The viscous and Maxwell step responses each admit a natural non-dimensionalization that reveals the governing control parameters and delineates the physically distinct relaxation regimes.

3.1.1 Viscous case: a universal single-parameter curve

Introducing the dimensionless distance $\hat{x} = x/(at)$ (the spatial coordinate measured in units of the instantaneous diffusion front at), the step displacement (17) collapses to

$$\frac{u^{\mathrm{STEP}}}{b} = \frac{1}{\pi}\arctan\left(\frac{1}{\hat{x}}\right). \tag{47}$$

Equation (47) is a universal curve: every viscous step response, regardless of the values of E, H, η, or b, maps onto this single function of $\hat{x}$. Near the rift axis or at late times ($\hat{x} \ll 1$), the displacement saturates at $u/b \to 1/2$; far from the rift or at early times ($\hat{x} \gg 1$), $u/b \approx 1/(\pi\hat{x}) \to 0$. The corresponding velocity profile is an equally compact Lorentzian,

$$\frac{at\, v^{\mathrm{STEP}}}{b} = \frac{1}{\pi}\,\frac{\hat{x}}{1+\hat{x}^2}, \tag{48}$$

with a peak located precisely at the diffusion front ($\hat{x} = 1$, i.e. $x = at$), an amplitude that decays as $b/(at)$, and a width that broadens linearly in time. The single parameter a thus encodes the entire spatio-temporal evolution of the post-diking transient.

The surface profile is only the boundary trace of a flow that fills the underlying half-space. Depth is reintroduced through the same harmonic extension used to construct the fractional Laplacian in Section 2.2: at depth z, the horizontal and vertical interior velocity components are the Poisson-kernel convolution of the surface velocity $V(x)$,

$$v_x(x,z) = \frac{2}{\pi}\int_{-\pi/2}^{\pi/2} \sin^2\theta\, V(x - z\tan\theta)\, d\theta, \qquad v_z(x,z) = \frac{2}{\pi}\int_{-\pi/2}^{\pi/2} \sin\theta\cos\theta\, V(x - z\tan\theta)\, d\theta, \tag{49}$$

with V set to the viscous step velocity $v^{\mathrm{STEP}}(x,t)$ or the viscous CSS velocity $v_{\mathrm{CSS}}(x,\tau)$ as appropriate. Figure 4 shows both, for a representative set of rheological parameters, at six illustrative phases τ spanning early (elastic-like) to late (diffused) states: the surface profile (top row of each block) alongside the interior flow it drives (bottom row, arrows coloured and scaled by speed). Both blocks are normalised by the same reference velocity V, so arrow colour is directly comparable between STEP and CSS. In both models, near-field flow is steep, shallow, and directed away from the rift at early τ; as τ grows, the same total flux spreads over a broadening and deepening lobe, exactly the widening self-similar front of Equations (47)–(48). The STEP panels show the true decay of the transient, with the peak surface velocity falling steadily from the shortest to the longest τ shown. The CSS interior field (bottom block) never empties out this way: it saturates on the far-field plate rate V by construction (Section 2.4), while the STEP field (top block) keeps thinning toward zero.

3.1.2 *Maxwell case: two dimensionless groups and three regimes*

The Maxwell step solution introduces a second dimensionless group. For positive wavenumbers $k > 0$, writing the Fourier-space exponent (34) as

$$-\frac{akt}{1+\tau_R ak} = -\frac{\hat{t}\,\kappa}{1+\kappa}, \tag{50}$$

where $\hat{t} = t/\tau_R$ is the Maxwell number and $\kappa = \tau_R ak$ is the wavenumber scaled by the Maxwell length

$$\ell_M := a\tau_R = \frac{EH}{2\mu_m(1-\nu^2)}, \tag{51}$$

three physically distinct relaxation regimes emerge. (i) *Elastic rebound* ($\kappa \gg 1$, i.e. $x \ll \ell_M$): the exponent in (50) saturates at $-\hat{t}$, independent of wavenumber, so near-field deformation decays at the uniform rate $1/\tau_R$, pure Maxwell relaxation of the mantle spring, independent of the elastic crust. (ii) *Viscous diffusion* ($\hat{t} \gg 1$ or $x \gg \ell_M$, $\kappa \ll 1$): the denominator $1+\kappa \approx 1$ and the Maxwell solution converges pointwise to the purely viscous step response of Section 2.3. (iii) *Crossover* ($\hat{t} \sim 1$): both timescales are simultaneously active; near-field deformation lags behind the viscous prediction (a short-time stiffening) before relaxing toward the common Cyclo-Steady State at $t \gg \tau_R$. The Maxwell length $\ell_M = EH/[2\mu_m(1-\nu^2)]$ is a *purely elastic* quantity, independent of viscosity: it measures the ratio of lithospheric to mantle elastic stiffness times the plate thickness, and sets the spatial boundary between the two asymptotic regimes.

Regime (i) admits a closed-form limit that sharpens this picture. As $\kappa \to \infty$ the exponent in (50) saturates at $-\hat{t}$ independently of k, so the high-wavenumber tail of the Maxwell velocity integral (37) reduces to the same Dirichlet kernel $\sin(kx)/k$ used to derive the viscous step response (16), now weighted by the x-independent factor $e^{-\hat{t}}/(\tau_R a)$. Evaluating that Dirichlet integral gives the near-field plateau

$$v_x^{\mathrm{MAX}}(x,t) \;\to\; \frac{b}{2\tau_R}\, e^{-t/\tau_R} \;=\; \frac{b}{t}\,\frac{\hat{t}}{2}\, e^{-\hat{t}}, \tag{52}$$

an x-independent value rather than a smoothly rising-then-decaying Lorentzian peak: at fixed t, near-axis velocity saturates at a level set purely by how much elastic relaxation has occurred, not by distance from the rift. Equation (52) is maximised at $\hat{t} = 1$, i.e. when the post-diking observation time equals the Maxwell relaxation time. For $\hat{t} \gg 1$ (fast-relaxing mantle, $\tau_R \ll t$) the elastic phase has already decayed away by the time of observation and the plateau vanishes; for $\hat{t} \ll 1$ (slow-relaxing, near-elastic mantle) the spring has barely begun to release and the plateau is again small. Because the plateau is independent of x while the viscous Lorentzian (48) rises from zero at the rift axis, the two curves necessarily cross at some $x_c(\tau_R) < \ell_M$: Maxwell velocity exceeds the viscous prediction for $x < x_c$ and falls below it for $x_c < x < \ell_M$. The crossover moves toward the rift axis as τ_R increases, so the near-field enhancement of (52) is confined to a shrinking neighbourhood of the rift for an increasingly elastic mantle.

Figure 5 illustrates this behaviour for the step response. Both the snapshot time t and the Maxwell length $\ell_M = a\tau_R$ are anchored to the shared-modulus-closure relaxation time of Equation (46), evaluated at the same (Natron best-fit) a used for every curve in the figure – not an additional assumption, but the one τ_R already singled out by that closure for this a. Seven Maxwell relaxation times spanning two orders of magnitude around this closure value (from one tenth to ten times it) are plotted against x/ℓ_M on a logarithmic axis to resolve the near-field structure. Each Maxwell curve flattens into the plateau of (52) well inside $x < \ell_M$, crosses the purely viscous Lorentzian at the marked point x_c, and converges onto it in the far field. The plateau height is non-monotonic in τ_R, peaking when τ_R matches the closure value and decaying for both smaller and larger τ_R, consistent with (52); the crossover moves from $x_c \approx 0.92\,\ell_M$ at the smallest τ_R shown to $x_c \approx 0.14\,\ell_M$ at the largest. Figure 6 traces the cumulative displacement at $x = \ell_M$ through the crossover: Maxwell models lag the viscous prediction for $t \lesssim \tau_R$ before converging to the common asymptote $b/2$. Together, the two

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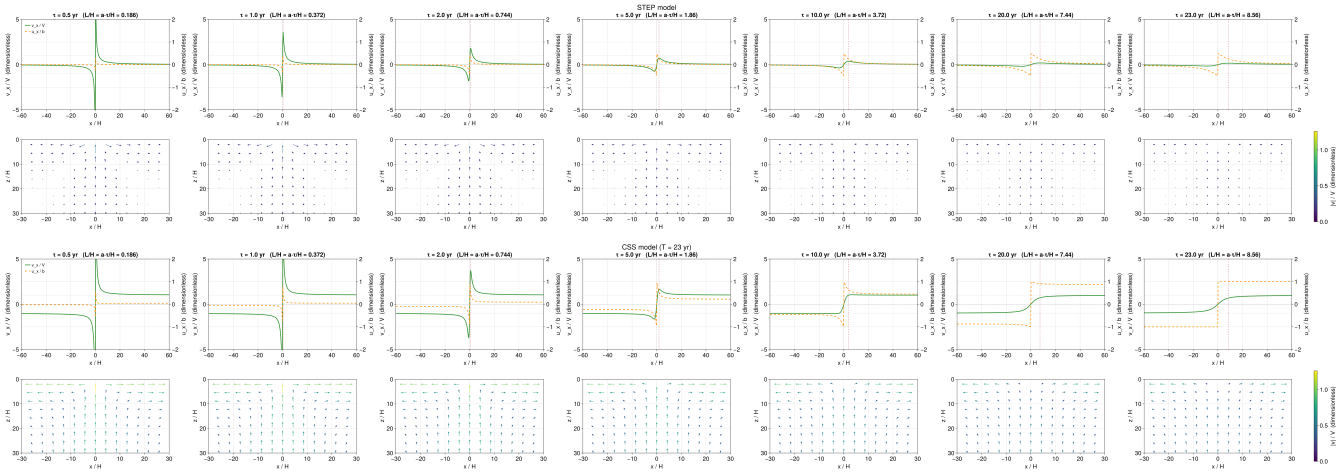


Figure 4. Surface velocity profile (top row of each block) and interior velocity field (bottom row; arrows coloured by speed magnitude, via Equation 49) for the viscous STEP response (top two rows) and the viscous CSS (bottom two rows), at six illustrative recurrence times τ. Both rows are normalised by the same reference velocity V (the plate rate implied by the CSS cycle period), so arrow colour is directly comparable between STEP and CSS; arrow length is scaled independently within each block for visual clarity. The dashed vertical line in each surface-profile panel marks the diffusion front $L = a\tau$. The single grid point directly above the dike is omitted from the quiver because the near-source field there is disproportionately large relative to the rest of the domain.

figures define an observational criterion. Post-diking geodetic records collected within a few τ_R of the rifting event, and within the shrinking near-axis zone $x < x_c(\tau_R)$, sit in the elastic-rebound regime, where the purely viscous step response underestimates near-field velocity; beyond x_c but still inside ℓ_M, the same viscous model overestimates it. Once $t \gg \tau_R$ the elastic spring has fully relaxed and the simpler viscous model is an accurate substitute everywhere.

3.1.3 Cyclo-Steady State: the deformation-width scaling

The CSS introduces a different natural length scale. In the viscous CSS (22), the Digamma argument is $\tau/T + ix/(aT)$, so the deformation width scales as aT rather than at. Defining $\hat{x}_{\rm CSS} = x/(aT)$ and $\hat{\tau} = \tau/T \in [0, 1)$, the CSS velocity collapses to

$$\frac{T\, v_{\rm CSS}}{b} = \frac{2}{\pi}\, \mathrm{Im}\{\psi(\hat{\tau} + i\hat{x}_{\rm CSS})\}\,, \qquad (53)$$

a universal function of $(\hat{\tau}, \hat{x}_{\rm CSS})$ alone. Unlike the step response, the CSS velocity profile does not broaden with time: its width is locked at aT by the periodicity of the forcing. For the Maxwell CSS (42), the ratio τ_R/T enters as a third dimensionless group. When $\tau_R/T \ll 1$ the Maxwell CSS converges pointwise to (53); when $\tau_R/T \sim 1$, near-field stiffening persists into the steady state, narrowing the velocity peak relative to the purely viscous prediction (Figure 5).

Figure 7 extends this comparison to the interior flow field, mirroring Figure 4 but for the Maxwell rheology: velocities are normalised by τ_R/b rather than $1/V$, and the fixed Maxwell length ℓ_M (Equation 51) replaces the growing diffusion front $a\tau$ as the vertical guide in every panel. The Maxwell STEP interior field (top two rows) inherits the near-field plateau of Equation (52): peak surface velocity is largest at the shortest τ shown and falls with increasing τ, and the underlying flow lobe shrinks and thins in step, the same qualitative decay as the viscous STEP field but paced

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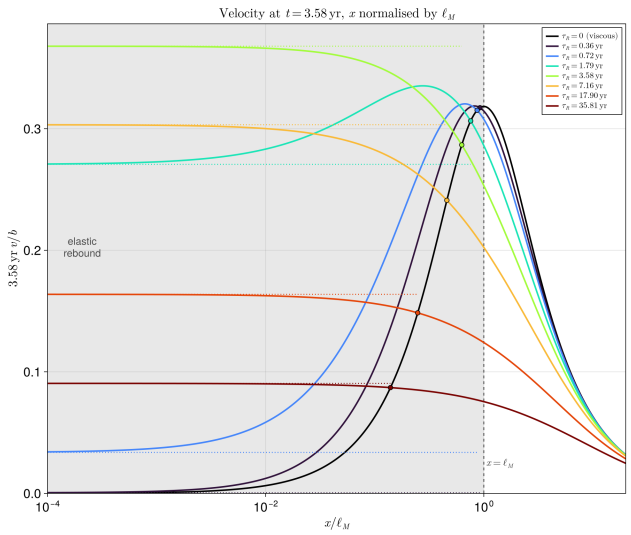


Figure 5. Maxwell visco-elastic near-field velocity, at the snapshot time and for the Maxwell length $\ell_M = a\tau_R$ set by the shared-modulus-closure relaxation time of Equation (46) at the Natron best-fit a – not an independently assumed value, but the one already singled out by that closure for this a. Distances are normalised by ℓ_M, velocities by b/τ_R evaluated at that same closure value. Grey shading marks the elastic-rebound zone ($x < \ell_M$). Seven values of τ_R, spanning one tenth to ten times the closure value, plus the purely viscous limit (black), which peaks at $x = \ell_M$ (dashed line). Each Maxwell curve flattens into the x-independent near-field plateau of Equation (52) (dotted, same colour), crosses the viscous Lorentzian at the marked point $x_c(\tau_R)$, and converges onto it in the far field. The plateau height is largest when τ_R matches the closure value and decreases for τ_R away from it in either direction, while x_c decreases monotonically as τ_R increases.

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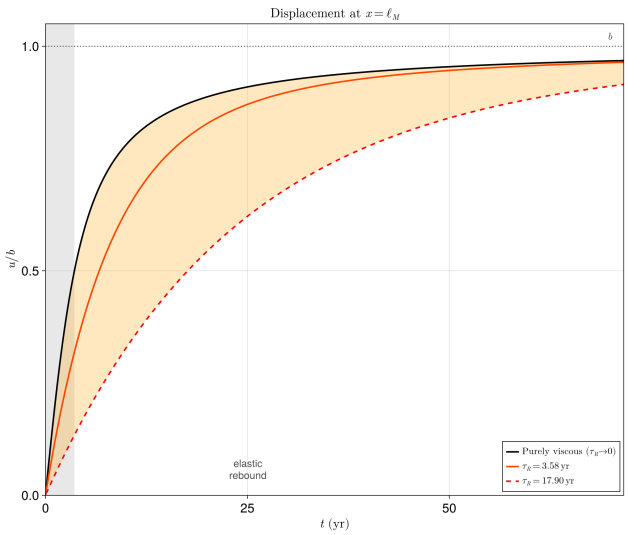


Figure 6. Maxwell visco-elastic cumulative displacement. Displacement is normalised by b. Grey shading marks the elastic-rebound zone (t below the shared-modulus-closure relaxation time of Equation 46 at the Natron best-fit a). Cumulative displacement at $x = \ell_M$ for the viscous model (black) and two Maxwell cases (τ_R set to that closure value, orange; to five times it, dashed red). Orange shading marks the displacement deficit relative to viscous. Both Maxwell models converge to the common asymptote $b/2$ (dotted) after $t \gg \tau_R$.

by τ_R rather than by the diffusion front. The Maxwell CSS interior field (bottom two rows) traces the same evolution within a single recurrence interval: immediately after an intrusion (τ small) the profile is sharply peaked, the elastic-rebound signature of Equation (52); as τ grows toward T the peak broadens and flattens, the spring relaxing toward the smoother, viscous-like profile it would reach were no further dike to intervene. This within-cycle relaxation is the spatial counterpart of the near-field stiffening described above ($\tau_R/T \sim 1$): the Maxwell CSS never fully forgets the elastic response of its most recent intrusion, in contrast with the viscous CSS, which saturates cleanly on the far-field plate rate throughout the cycle (Figure 4).

At each site, model parameters are estimated by minimizing the chi-squared misfit

$$\chi^2 = \sum_{i=1}^{N} \left(\frac{u_i^{\mathrm{obs}} - u_i^{\mathrm{pred}}}{\sigma_i} \right)^2 , \tag{54}$$

where u_i^{obs} and u_i^{pred} are the observed and predicted displacement (or baseline change) at data point i, σ_i is the 1σ measurement uncertainty, and N is the number of observations. Goodness of fit is quantified through the reduced statistic $\chi_r^2 = \chi^2/(N-p)$, where p is the number of free parameters; $\chi_r^2 \approx 1$ indicates that the model reproduces the data within the measurement noise.

3.2 Krafla rift, Iceland

The 1984 Krafla rifting episode in NE Iceland provides one of the best-documented post-diking geodetic records available. Foulger et al. (1992) measured horizontal surface displacements at 20 stations across the rift zone at two epochs 8 and 11 years after the 1975–1984 rifting sequence. We fit the differenced displacement field $\Delta u = u(\tau_f) - u(\tau_i)$, with $\tau_i = 8$ yr and $\tau_f = 11$ yr, using the CSS forward models of Sections 2.4 and 2.6: the viscous CSS, with free parameters (a, b) at fixed recurrence interval T and with T itself free; and its Maxwell counterpart, with τ_R derived from a through the shared-modulus closure (46) rather than fit independently, so that each Maxwell variant carries exactly as many free parameters as its viscous counterpart. Parameters are estimated by minimizing equation (54) over a Latin Hypercube sample in log-uniform parameter space, followed by bounded Nelder-Mead refinement.

Leaving T free returns $T \approx 3800$–4300 yr for the viscous and Maxwell CSS respectively – one to two orders of magnitude longer than the ~ 250 yr gap between the documented 1724–1729 and 1975–1984 eruptive crises at Krafla (Section 4.2). This is not a measurement of the true recurrence interval: at 8–11 yr post-rifting the observations span only a tiny fraction of any plausible T, so the fit is free to push T arbitrarily large, toward the single-event limit in which the CSS reduces to the step response of Section 2.3. We therefore also fix $T = 200$ yr, consistent with the documented recurrence history, as the physically anchored comparison.

All four CSS variants fit the observed displacement profile within the measurement noise (Figure 8, Table 1). The viscous CSS gives $a = 5.17 \times 10^{-5}$ m s^{-1}, $b = 0.59$ m ($\chi_r^2 = 0.72$) at $T = 200$ yr, and $a = 4.87 \times 10^{-5}$ m s^{-1}, $b = 0.59$ m, $T = 4289$ yr ($\chi_r^2 = 0.76$) with T free – both close to the best-fit characteristic velocity obtained from the step response alone, as expected since $\tau_i, \tau_f \ll T$ in every variant. The Maxwell CSS gives $a = 1.42 \times 10^{-4}$ m s^{-1}, $b = 0.50$ m, $\tau_R = 2.99$ yr (derived) at $T = 200$ yr ($\chi_r^2 = 0.78$), and $a = 1.34 \times 10^{-4}$ m s^{-1}, $b = 0.46$ m, $T = 3789$ yr, $\tau_R = 3.15$ yr (derived) with T free ($\chi_r^2 = 0.83$). Because each Maxwell fit is compared against a viscous fit with the same number of free parameters, the AIC comparison reduces to a direct χ^2 contrast: $\Delta\mathrm{AIC} = 1.1$ ($T = 200$ yr) and 1.2 (T free), in both cases favouring the viscous CSS but far short of the $|\Delta\mathrm{AIC}| > 2$ threshold for positive evidence. As at the single-event step response (Section 2.3), the 8–11 yr Krafla record does not require a Maxwell asthenosphere: whatever elastic contribution the mantle half-space might make relaxes faster than the shortest interval sampled here. Using the viscous CSS best-fit a ($T = 200$ yr) in (7) yields $\eta = EH/[2a(1-\nu^2)] \approx 4.1 \times 10^{18}$ Pa s for canonical Icelandic lithospheric parameters ($H = 10$ km, $E = 40$ GPa, $\nu = 0.25$), consistent with post-glacial rebound estimates (Sigmundsson et al. 2010) and below the upper bound $\eta < 10^{19}$ Pa s placed by Sigmundsson (1991) for a 10-km elastic layer at Krafla.

Figure 9 shows the χ^2 confidence region for the viscous CSS (a, b) at $T = 200$ yr, the best-fitting model at this site. The 1σ contour is broad and elongated along an anti-correlated a–b trade-off: a larger characteristic velocity can be compensated by a smaller dike half-opening without degrading the fit, reflecting the modest station coverage of the single differenced epoch pair.

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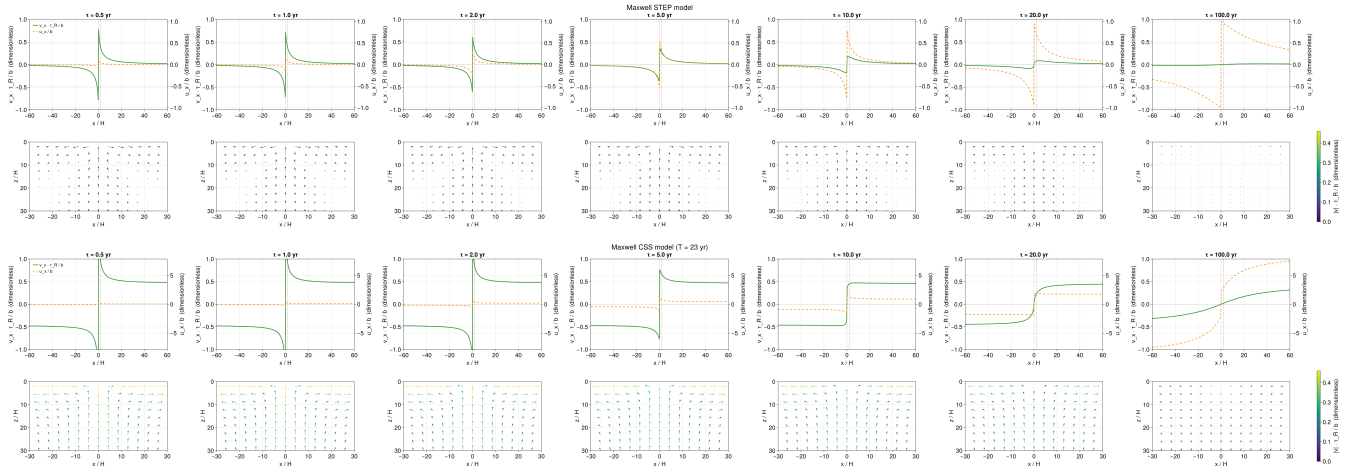


Figure 7. Surface velocity profile (top row of each block) and interior velocity field (bottom row; arrows coloured by speed magnitude, via Equation 49) for the Maxwell STEP response (top two rows) and the Maxwell CSS (bottom two rows), at six illustrative recurrence times τ. Both rows are normalised by τ_R/b (the Maxwell relaxation time), on the same colour scale so arrow colour is directly comparable between STEP and CSS; arrow length is scaled independently within each block for visual clarity. The dashed vertical line in each surface-profile panel marks the fixed Maxwell length $\ell_M = a\tau_R$ (Equation 51), rather than the viscous $L = a\tau$, since ℓ_M does not depend on τ. The single grid point directly above the dike is omitted from the quiver because the near-source field there is disproportionately large relative to the rest of the domain.

Table 1. CSS model comparison for the Krafla displacement profile ($N = 20$). MaxCSS carries τ_R derived from a (46), not as an independent free parameter, so k matches the corresponding viscous CSS row.

Model	k	χ^2_r	ΔAIC vs. viscous
CSS (T free)	3	0.76	–
CSS ($T = 200$ yr)	2	0.72	–
MaxCSS (T free)	3	0.83	+1.2
MaxCSS ($T = 200$ yr)	2	0.78	+1.1

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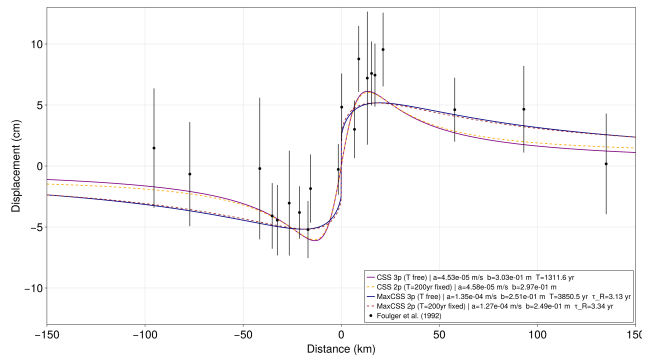


Figure 8. Best-fit CSS differenced displacement profiles for the Krafla post-diking dataset of Foulger et al. (1992). Observed rift-perpendicular displacements between $\tau_i = 8$ yr and $\tau_f = 11$ yr post-rifting are shown as filled circles with 1σ error bars (cm). Curves show the viscous CSS (T free, purple; $T = 200$ yr, dashed orange) and Maxwell CSS (T free, navy; $T = 200$ yr, dashed brown), with τ_R derived from a via Equation (46); see Table 1 and text for parameter values and goodness-of-fit statistics.

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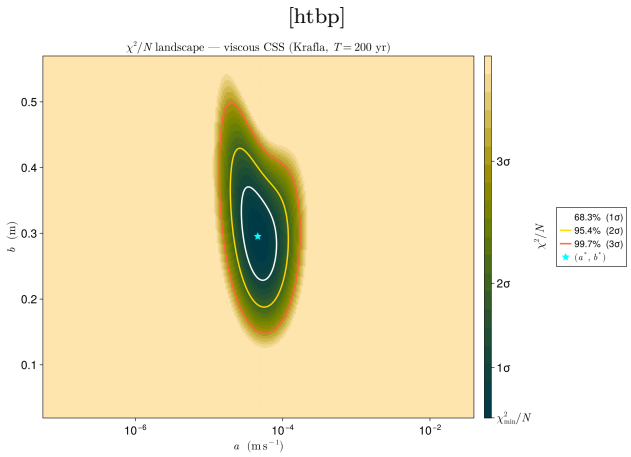


Figure 9. χ^2/N confidence region for the viscous CSS fit to the Krafla displacement profile ($T = 200$ yr, Table 1). Contours mark the 1σ, 2σ, and 3σ confidence levels; the star marks the best fit (a^*, b^*).

3.3 Asal rift, Djibouti

The November 1978 Asal–Ghoubbet rifting episode in the Afar Triangle opened approximately 2 m of rift-perpendicular extension over several days (Le Dain et al. 1979; Ruegg et al. 1984), launching one of the longest continuously monitored post-diking transient records outside Iceland. Vigny et al. (2007) tracked the EP00–DF00 baseline (a rift-perpendicular geodetic tie crossing the Asal rift) from 1979 to 2003, recording ≈ 490 mm of cumulative lengthening, with rapid early growth decelerating sharply after the mid-1980s. We digitize all 33 data points from Figure 3 of Vigny et al. (2007), combining trilateration campaigns (1979–1987) and GPS surveys (1991–2003), and fit the same four

CSS variants as at Krafla (Section 3.2): viscous CSS with T free or fixed, and its Maxwell counterpart with τ_R derived from a through (46). The predicted baseline change is evaluated as $\Delta L = 2u(x_0, \tau)$ at $x_0 = 2000$ m, half the EP00–DF00 rift-perpendicular separation, with the rifting onset fixed at $t_0 = 1978.85$ yr (Vigny et al. 2007). Because the Vigny et al. time series is referenced to the first measurement epoch ($\tau_{\rm ref} \approx 1.1$ yr after the rifting event) rather than to the rifting onset, we subtract from each model curve its own prediction at $\tau_{\rm ref}$, anchoring $\Delta L = 0$ at the reference epoch without introducing a free parameter. No documented rifting episode preceded 1978 at Asal, so we adopt $T = 500$ yr as the physically anchored fixed-T comparison, following the same reasoning as at Krafla.

All four CSS variants reproduce the overall shape of the Asal time series (Figure 10, Table 2), but the viscous CSS fits markedly better than its Maxwell counterpart at equal parameter count. The viscous CSS gives $a = 9.47 \times 10^{-6}$ m s^{-1}, $b = 0.354$ m ($\chi^2_r = 1.54$) at $T = 500$ yr, and $a = 9.70 \times 10^{-6}$ m s^{-1}, $b = 0.354$ m, $T = 2085$ yr ($\chi^2_r = 1.54$) with T free – as at Krafla, the free-T fit drifts far above any plausible recurrence interval, confirming that the 25-yr Asal record cannot constrain T independently of the single-event limit. The Maxwell CSS gives $a = 8.08 \times 10^{-5}$ m s^{-1}, $b = 0.316$ m, $\tau_R = 5.23$ yr (derived) at $T = 500$ yr ($\chi^2_r = 2.61$), and $a = 7.63 \times 10^{-5}$ m s^{-1}, $b = 0.325$ m, $T = 4066$ yr, $\tau_R = 5.54$ yr (derived) with T free ($\chi^2_r = 2.48$). Unlike at Krafla, the AIC comparison here is decisive: ΔAIC $= 33.4$ ($T = 500$ yr) and 28.1 (T free), both strongly favouring the viscous CSS ($|\Delta{\rm AIC}| \gg 6$). The 25-yr Asal baseline, unlike the single 8–11 yr Krafla epoch pair, spans enough of the post-diking transient that a spurious elastic contribution from the derived $\tau_R \approx 5$–6 yr measurably worsens the fit at late times, where the data are already well described by pure viscous relaxation.

Independent field observations, which favoured the viscous single-event response over the Foulger et al. (1992) diffusion model in earlier analyses of this same record, do not resolve as cleanly for the CSS. The viscous CSS half-opening $b = 0.354$ m implies a total dike opening $2b = 0.71$ m, and the Maxwell CSS gives $2b = 0.63$ m; both fall further below the ≈ 1.9 m measured across the rift by trilateration immediately after the 1978 event (Ruegg et al. 1979) than a single-event fit unconstrained by the rest of the 25-yr record. The same pattern appears in the early deformation rate: the viscous CSS predicts ≈ 33 mm yr^{-1} at $\tau_{\rm ref} \approx 1.1$ yr and the Maxwell CSS ≈ 39 mm yr^{-1}, both below the ≈ 53 mm yr^{-1} measured by trilateration shortly after the event (Ruegg & Kasser 1987). This is a consequence of the CSS parameterisation itself: tying the plate rate to $v_{\rm plate} = b/T$ links b to the late-time behaviour of the whole 25-yr record, not only to its rapid early rise, and pulls the best-fit b below the value an early-epoch-only fit would return. The viscous best-fit $a = 9.47 \times 10^{-6}$ m s^{-1} nonetheless implies a sub-lithospheric viscosity of $\eta \approx 2.3 \times 10^{19}$ Pa s for Afar lithospheric parameters ($H = 10$ km, $E = 40$ GPa, $\nu = 0.25$), in the range of estimates from other post-diking transient analyses in the East African Rift (Nooner et al. 2009).

Vigny et al. (2007) also measured rift-perpendicular GPS velocities along a transect crossing the rift at a mean epoch of ≈ 19 yr after the 1978 event. We forward-predict the velocity profile from the four CSS time-series best fits, placing the model centre at $x = +1$ km northeast of the topographic rift axis to coincide with the Petit Rift (Vigny et al. 2007). All four variants reproduce the near-rift velocity gradient (Figure 11). The dominant misfit is the ≈ 16 mm yr^{-1} Somalia–Danakil background plate motion (Saria et al. 2014; Stamps et al. 2021), which produces a pronounced SW–NE velocity asymmetry absent from our post-rift transient models; we fit a uniform background velocity ($V_{\rm plate} \approx 8$ mm yr^{-1} in every variant) to each profile before comparing shapes. The viscous CSS again fits better than its Maxwell counterpart, and slightly better than the single-event step response reported in earlier analyses of this profile: $\chi^2_r = 15.7$ (T free) and 15.9 ($T = 500$ yr) for the viscous CSS, versus 22.6 (T free) and 22.2 ($T = 500$ yr) for the Maxwell CSS. None of the four variants reaches $\chi^2_r \sim 1$; as before, the elevated residuals reflect the asymmetric far-field Somalia–Danakil signal that a single symmetric transient and a uniform velocity offset cannot fully capture, not a failure specific to either rheology.

Figure 12 shows the χ^2 confidence region for the viscous CSS (a, b) at $T = 500$ yr, the best-fitting model at this site. Unlike Krafla, the 25-yr time series confines (a, b) to a narrow, strongly anti-correlated ellipse: the longer baseline resolves the characteristic velocity and dike half-opening far more tightly, which is precisely why the Maxwell CSS is decisively rejected here (Table 2) but not at Krafla.

Table 2. CSS model comparison for the Asal EP00–DF00 time series ($N = 33$). MaxCSS carries τ_R derived from a (46), matching k to the corresponding viscous CSS row.

Model	k	χ^2_r	ΔAIC vs. viscous
CSS (T free)	3	1.54	–
CSS ($T = 500$ yr)	2	1.54	–
MaxCSS (T free)	3	2.48	+28.1
MaxCSS ($T = 500$ yr)	2	2.61	+33.4

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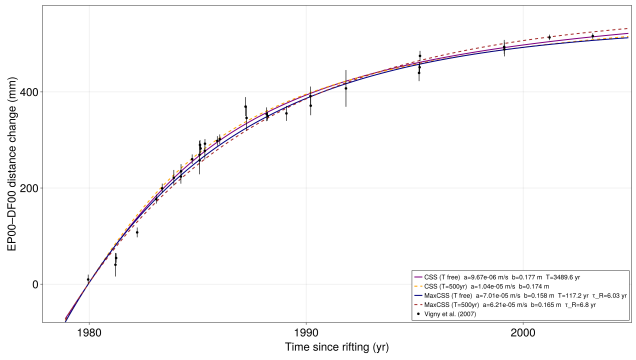


Figure 10. Best-fit CSS forward models for the Asal post-diking time series of Vigny et al. (2007). The EP00–DF00 baseline change (mm) is shown as filled circles with 1σ error bars. Curves show the viscous CSS (T free, purple; $T = 500$ yr, dashed orange) and Maxwell CSS (T free, navy; $T = 500$ yr, dashed brown), with τ_R derived from a via Equation (46). All curves are referenced to the first measurement epoch ($\tau_{\rm ref} \approx 1.1$ yr); see Table 2 and text for parameter values and goodness-of-fit statistics.

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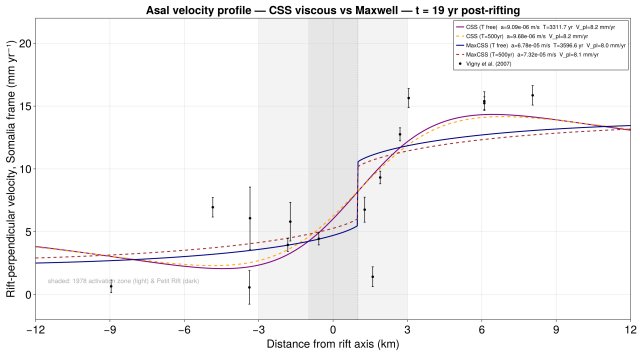


Figure 11. Forward-predicted rift-perpendicular velocity profiles at $t \approx 19$ yr post-rifting, compared with GPS observations of Vigny et al. (2007) (Somalia reference frame, mm yr^{-1}). Curves show the viscous CSS (T free, purple; $T = 500$ yr, dashed orange) and Maxwell CSS (T free, navy; $T = 500$ yr, dashed brown), with parameters fixed at the time-series best-fit values (Figure 10) and a uniform plate-motion offset ($V_{\rm plate} \approx 8$ mm yr^{-1}) added to each curve. Shading marks the 1978 activated zone (±3 km, light) and the Petit Rift (±1 km, dark); the dashed vertical line marks the model centre at $x = +1$ km.

4 DISCUSSION

4.1 A boundary-condition-free description of inter-rifting deformation

The CSS framework closes the kinematic gap identified in Section 1 without introducing a locking depth, a prescribed domain width, or an artificially imposed far-field velocity. Backslip models of inter-rifting deformation (Savage & Prescott 1978) and their dislocation equivalents (LaFemina et al. 2005; Konca et al. 2018) reproduce observed velocity gradients only at the cost of a free geometric parameter (the locking depth or creep-zone geometry) that carries no independent mechanical constraint in magmatic rifts where the rheological transition is gradational rather than sharp. Numerical visco-elastic simulations face an equivalent obstacle: in any finite domain, lateral boundary conditions must either force far-field velocities to zero, in contradiction with global plate motion models (Altamimi et al. 2016), or prescribe the divergence rate that the model is itself meant to predict (Hofton et al. 1995; Hofton & Foulger 1996).

The CSS requires neither. The deformation zone half-width $aT = EHT/[2\eta(1-\nu^2)]$ emerges directly from the balance between elastic restoring stresses in the plate and viscous drag at its base, parameterized entirely by the characteristic velocity a and the recurrence interval T. Far-field plate divergence is satisfied by construction through the conservation property of the Digamma representation (equation 24): no boundary condition is imposed, and no artificial velocity offset is needed to reconcile model predictions with far-field geodesy. Both free parameters are physically grounded: the dike half-opening b is measurable independently from field geology and geodetic surveys (Le Dain et al. 1979; Ruegg et al. 1984), and a directly constrains the ratio EH/η without requiring an independent estimate of viscous layer thickness. This structural simplicity distinguishes the CSS from existing kinematic and semi-analytical approaches to the inter-rifting problem.

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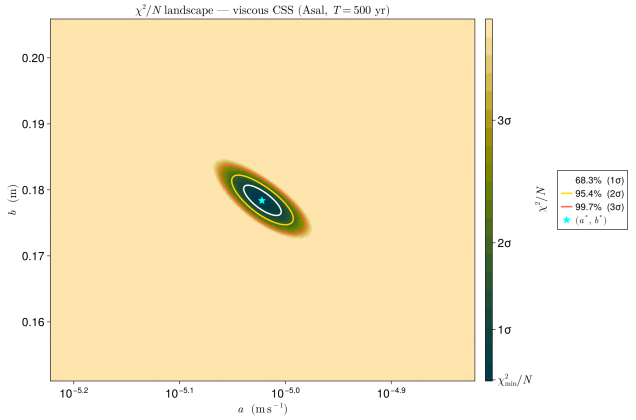


Figure 12. χ^2/N confidence region for the viscous CSS fit to the Asal time series ($T = 500$ yr, Table 2). Contours mark the 1σ, 2σ, and 3σ confidence levels; the star marks the best fit (a^*, b^*). Note the much narrower range spanned by both axes relative to Figure 9.

4.2 Rheological constraints and deformation zone scaling

The single composite parameter $a = EH/[2\eta(1-\nu^2)]$ collapses the elastic–viscous trade-off onto a one-dimensional family of solutions in (E, H, η) space. Figure 13 makes this trade-off explicit: for fixed E, the iso-a contour traces a band across (H, η) space. Dark shading propagates the 1σ uncertainty on a from the χ^2 fit at fixed $E = 40$ GPa; light shading additionally propagates E over [20, 80] GPa (combined 1σ envelope). The iso-a curves are nearly parallel in this representation, so the dominant uncertainty is in η rather than in H. Independent seismological constraints on the seismogenic layer thickness ($H \approx 5$–15 km in both Iceland and the Afar; Stevens et al. 2021; Plasman et al. 2017; Weinstein et al. 2017 break the remaining ambiguity and allow us to translate the geodetically constrained a into a viscosity estimate for each site (Figure 13).

Adopting $H = 10$ km and $E = 40$ GPa, the Krafla viscous-CSS best-fit $a = 5.17 \times 10^{-5}$ m s^{-1} ($T = 200$ yr, Section 3.2) implies $\eta \approx 4.1 \times 10^{18}$ Pa s, essentially the same estimate the single-event step response would give, since $\tau_i, \tau_f \ll T$ places the fit in the regime where the two forward models coincide. Two independent sources of uncertainty affect this estimate. The 1σ confidence interval on a from the χ^2 fit (Section 3.2) yields $\eta \in [2.6, 7.4] \times 10^{18}$ Pa s for fixed $E = 40$ GPa, while propagating E over [20, 80] GPa at the best-fit a gives $\eta \in [2.1, 8.3] \times 10^{18}$ Pa s. Because $\eta = EH/[2a(1-\nu^2)]$, the two uncertainties are additive in logarithmic space; combining them in quadrature ($\sigma_{\log_{10} E} = 0.30$, $\sigma_{\log_{10} a} = 0.23$) yields a combined 1σ range of $\eta \in [1.7, 10.0] \times 10^{18}$ Pa s (Figure 13, left). This range lies entirely below or at the upper bound $\eta < 10^{19}$ Pa s placed by Sigmundsson (1991) for a 10-km elastic layer at Krafla, and is consistent with post-glacial rebound estimates across Iceland ($\eta \sim 10^{18}$–10^{19} Pa s; Sigmundsson et al. 2010) and with forward visco-elastic modelling of the same post-diking transient (Hofton & Foulger 1996). The low viscosity beneath Iceland reflects anomalously high temperatures in the

sub-lithospheric mantle maintained by the Iceland plume: partial melt in the lower crust and uppermost mantle reduces the effective viscosity by one to two orders of magnitude relative to standard continental lithosphere (Holtzman 2016).

The Asal viscous-CSS best-fit $a = 9.47 \times 10^{-6}$ m s^{-1} ($T = 500$ yr, Section 3.3) implies $\eta \approx 2.3 \times 10^{19}$ Pa s at $H = 10$ km and $E = 40$ GPa, with an E-propagated range of $\eta \in [1.1, 4.5] \times 10^{19}$ Pa s, approximately five times higher than the Krafla estimate. This is higher than viscosity estimates for the Dabbahu segment of the same rift system ($\eta \approx 5 \times 10^{18}$ Pa s; Nooner et al. 2009; $\eta \sim 10^{18}$ Pa s, $H = 21$ km; Hamling et al., 2014 [CITE NEEDED]), reflecting either genuine segment-to-segment variation in sub-lithospheric temperature or contributions from shallow magmatic sources that the purely viscous CSS cannot distinguish from mantle relaxation. Smittarello et al. (2016) [CITE NEEDED] argue, from 38 years of geodetic observations of the Asal–Ghoubbet rift, that magma supply rather than viscoelastic relaxation controls the post-diking velocity field at this site, a degeneracy our single-parameter model cannot resolve without independent constraints on the magmatic budget. The factor-of-five viscosity contrast between Krafla and Asal nevertheless remains physically coherent: both are magmatically active rifts, but Iceland's plume-driven heat flux maintains substantially higher sub-lithospheric temperatures, accelerating viscous relaxation relative to the East African setting.

Figure 14 translates the best-fit a values into predictions for the CSS deformation zone width aT, which controls how broadly inter-rifting velocities are distributed across a geodetic network. For a Krafla recurrence interval of $T \approx 200$ yr, consistent with the ~ 250 yr gap between the 1724–1729 and 1975–1984 eruptive crises, the CSS width reaches $aT \approx 326$ km, already exceeding the 230 km span of the Foulger et al. (1992) network (Figure 14), consistent with the broad velocity gradients observed across Iceland (LaFemina et al. 2005). For Asal the recurrence interval is poorly constrained; adopting $T \approx 500$ yr gives $aT \approx 149$ km, nearly ten times the 17 km aperture of the Vigny et al. (2007) GPS profile (Figure 14). In both cases the CSS width vastly exceeds the network aperture, which is precisely why inter-rifting velocity fields in Iceland and the Afar have been interpreted as steady plate divergence rather than as the superposition of slowly decaying visco-elastic transients. This is also why the free-T fits of Sections 3.2 and 3.3 cannot recover T itself: once aT exceeds the network aperture by an order of magnitude, the data constrain only a, and χ^2 is essentially flat with respect to T beyond that point.

4.3 Non-local diffusion versus the Fickian model

The preference for the viscous CSS over the Foulger diffusion model is not a numerical accident: it reflects a structural difference in the governing equations. Foulger et al. (1992) model post-diking deformation as Fickian diffusion of stress, governed by $\partial u/\partial t = \kappa\, \partial^2 u/\partial x^2$, whose Fourier signature is $\partial \hat{u}/\partial t = -\kappa k^2 \hat{u}$. The quadratic wavenumber dependence concentrates decay at short wavelengths and generates an erfc displacement profile with a deformation front that propagates as $\sqrt{\kappa t}$ and leaves the far field exponentially unperturbed on timescales shorter than x^2/κ. Our

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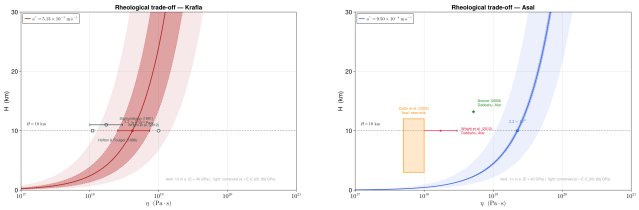


Figure 13. Rheological trade-off for Krafla (left, red) and Asal (right, blue), using the single-event step-response χ^2 landscape; the viscous-CSS best-fit a (Sections 3.2, 3.3) is statistically indistinguishable ($< 1\%$ different) at both sites, so the same iso-a curve and uncertainty envelope apply. Each panel shows the iso-a contour in the (H, η) plane at the best-fit value. Dark shading propagates the 1σ uncertainty on a from the χ^2 fit at fixed $E = 40$ GPa; light shading additionally propagates E over [20, 80] GPa (combined 1σ envelope). The $H = 10$ km dashed line marks the reference elastic thickness; markers show published viscosity estimates (Krafla: open circle, Sigmundsson 1991; open square, Hofton & Foulger 1996; open circle with error bar, Wright et al. 2012; Asal: filled diamond, Nooner et al. 2009 for the Dabbahu segment of the same rift system; orange box, Cattin et al. 2005 — near-axis effective viscosity $\eta = 0.5$–1×10^{18} Pa s from a 2D thermo-mechanical FEM model of the first 6–8 yr post-rifting at Asal, with the seismogenic layer thickness $H = 3$–12 km from the same study). The Cattin et al. box lies well below the iso-a^* curve, consistent with the local near-axis viscosity being lower than the laterally averaged bulk estimate from the 25-yr geodetic record.

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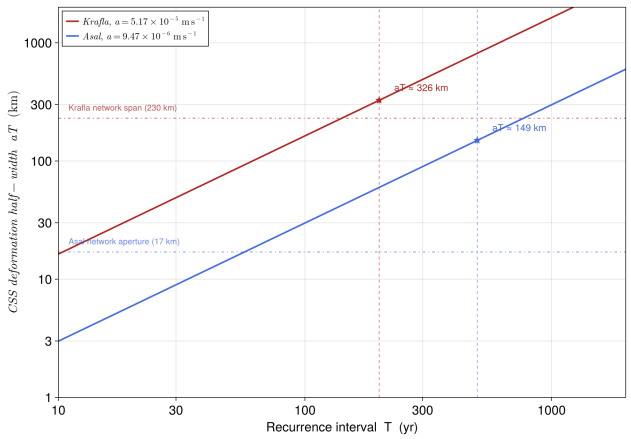


Figure 14. CSS deformation half-width aT as a function of recurrence interval T for Krafla (red, $a = 5.17 \times 10^{-5}$ m s^{-1}) and Asal (blue, $a = 9.47 \times 10^{-6}$ m s^{-1}). Vertical dashed lines mark estimated recurrence intervals; annotated points give the implied aT values. Horizontal dash-dot lines mark the total span of each geodetic network used in this study: 230 km for Krafla (Foulger et al. 1992, 1992; distance between outermost stations) and 17 km for Asal (Vigny et al., 2007; GPS profile extent). In both cases aT substantially exceeds the network aperture, explaining why inter-rifting deformation has been interpreted as steady plate divergence rather than a broad visco-elastic transient.

viscous model is governed instead by (8), $\partial\hat{u}/\partial t = -a|k|\hat{u}$, where the absolute value $|k|$ is the Fourier representation of the half-Laplacian $(-\partial^2/\partial x^2)^{1/2}$, a spatially *non-local* operator arising from the $1/(x-x')^2$ Stokes convolution kernel (3). Because the mantle transmits stress instantaneously through the incompressible Stokes flow, every point on the elastic plate feels the rifting event from the first instant; no diffusive front is required to reach it. The surface velocity is therefore a Lorentzian (48), with tails decaying as $\sim 1/x^2$ rather than the exponential or Gaussian decay of the Fickian model.

The observational consequence is directly visible in the Asal velocity transect (Figure 11). At stations several kilometres from the rift axis (beyond the zone where model differences are masked by near-field noise), the viscous CSS fits the data better than the Foulger model ($\chi^2_r \approx 15.7$–15.9 vs 33, better than a factor-of-two improvement), because the Lorentzian distributes deformation more broadly than the erfc profile. The same asymmetry drives the profile misfit more clearly at Asal than at Krafla, where the differenced-displacement observable $\Delta u = u(\tau_f) - u(\tau_i)$ suppresses far-field signal regardless of the spatial functional form. The non-local character of the Stokes kernel thus has a concrete, testable imprint on the spatial pattern of inter-rifting deformation: networks sampling distances $x \gg aT/\pi$ are more sensitive to the tail behaviour and therefore better placed to discriminate between the two formulations.

The two formulations are not independent alternatives, however: the Foulger diffusion profile is mathematically embedded within the viscous solution. Bochner's subordination formula writes the half-Laplacian semigroup as a weighted average, over all internal diffusion times, of the ordinary Fickian heat semigroup (Metzler & Klafter 2000). Substituting the model's own initial condition into this average reproduces the closed-form viscous step response exactly. The Foulger model is therefore not simply the wrong physical model: it is the correct solution evaluated at a single diffusion timescale, rather than blended across all of them.

4.4 Robustness under stochastic recurrence

The CSS derivation assumes strictly periodic rifting: intrusions occur at fixed interval T and deliver a fixed half-opening b at each event. Neither assumption holds in nature. Magmatic rifting sequences are episodic and irregular; the $\sim$ 250-year gap between the Krafla 1724–1729 and 1975–1984 crises is a single estimate from a sparse historical record, and dike opening magnitudes vary substantially within and between sequences (Thordarson & Larsen 2007). The CSS remains a robust estimator of the long-term inter-rifting velocity field even under substantial cycle-to-cycle variability in timing and dike opening, as we demonstrate through stochastic ensembles.

We replaced the deterministic event sequence with stochastic ensembles in which each past rifting event draws its recurrence interval T_j and half-opening b_j independently from a Normal distribution with mean (T, b) and coefficient of variation $c_v = 0.30$, a scatter representative of documented variability in Icelandic and Afar rift chronologies. For each of 500 realisations we summed the step-response contributions from all past events and extracted the 10th–90th percentile band across the ensemble (Figure 15).

Four results stand out. First, the ensemble median coincides with the deterministic CSS in all four cases: linearity of the step-response sum and the conditions $\mathbb{E}[T_j] = T$ and $\mathbb{E}[b_j] = b$ guarantee that the ensemble average recovers the periodic solution exactly, regardless of the scatter level. Second, the spread differs sharply between the two sources of variability. Scatter in T (panel a) concentrates uncertainty in the transition zone near $\hat{x} \sim 1$: irregular timing shifts the velocity gradient left or right across realisations, smearing the transition zone while the far-field plateau $v \to b/T$ remains sharp. Scatter in b (panel b) acts differently: because the recurrence interval is fixed, events arrive at the same times in every realisation and the profile shape is identical to the deterministic CSS in all of them; only its overall amplitude varies. The effect is equivalent to sliding each realisation's curve up or down the y-axis by a random factor, so the band at each distance is a fixed fraction of the local CSS value and neither the gradient position nor the far-field plateau is affected. Third, the combined scatter (panel c) produces the widest band; at the deformation-width scale $\hat{x} = 1$, the 10–90th percentile range spans roughly one-third of the median CSS amplitude for $c_v = 0.30$, and the band narrows steadily toward the far field as the law-of-large-numbers averaging over many past events takes effect. This width confirms that the CSS is a robust estimator of the time-averaged inter-rifting velocity field even under substantial cycle-to-cycle variability. Fourth, imposing the plate-rate-conserving coupling $b_j = v_{\mathrm{plate}} T_j$ (panel d) causes the far-field band to collapse to essentially zero: because $b_j/T_j = v_{\mathrm{plate}}$ for every event in every realisation, the plate velocity is encoded exactly regardless of timing scatter. Residual spread in the transition zone ($|\hat{x}| \lesssim 1$) persists because stochastic timing still shifts the position of the velocity gradient even though its far-field amplitude is constrained. This scenario corresponds physically to a self-regulating rift in which crustal opening scales with the elapsed time since the previous event, a behaviour consistent with the observation that magmatic rifts accommodate most of the long-term plate divergence over geological timescales (Thordarson & Larsen 2007).

The central-limit interpretation is the following. A geodetic snapshot of the inter-rifting velocity field at any site within the deformation zone $x \lesssim aT$ encodes superposed contributions from dozens of past events. Stochastic scatter in individual T_j and b_j averages down over the ensemble of contributing events, and the CSS prediction converges to within the GPS measurement uncertainty for networks with aperture comparable to aT. Near-rift stations that respond primarily to the single most recent event retain the full cycle-to-cycle scatter shown in panel (a), and the CSS approximation is less accurate there; in that regime, knowledge of the specific rifting history is required to reduce the uncertainty.

4.5 Perspectives: the CSS as an inversion tool and its extension to heterogeneous rifts

The applications presented here treat T and b as known inputs and fit the single free parameter a to geodetic data. The CSS framework permits a more ambitious inversion. Because the velocity profile $v_{\mathrm{CSS}}(x, \tau)$ depends on three quantities (the deformation zone half-width aT, the far-field plate ve-

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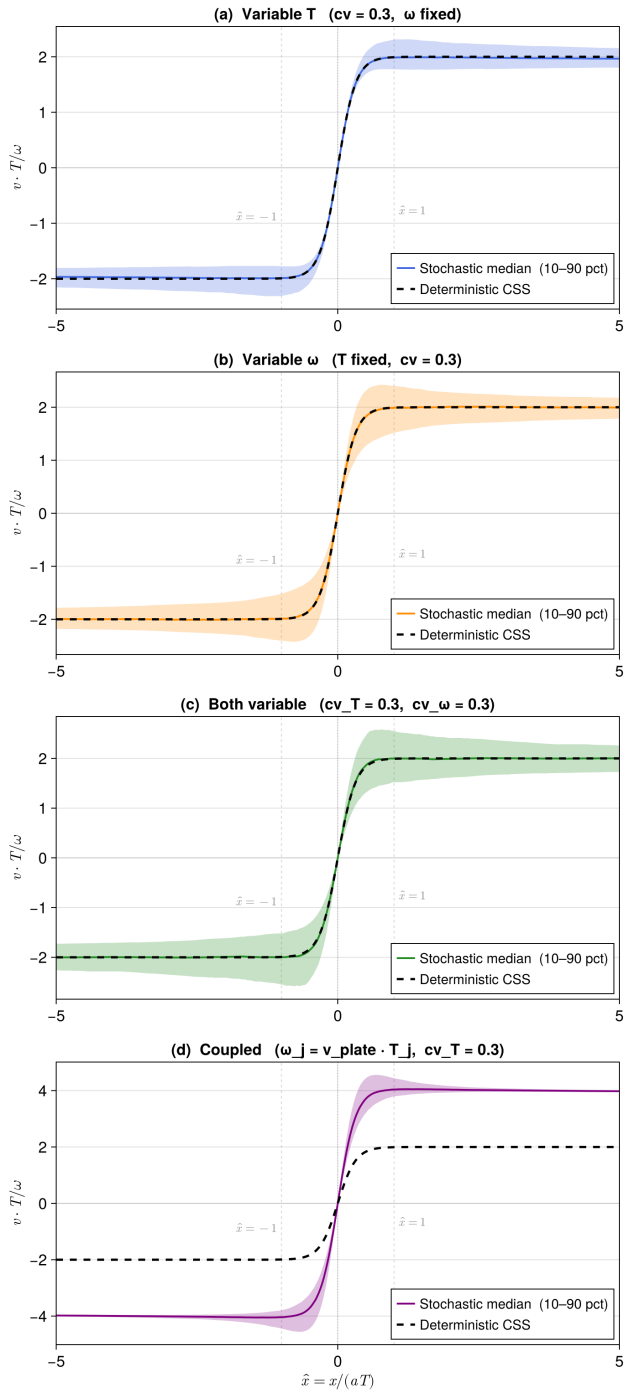


Figure 15. Robustness of the CSS velocity profile under stochastic recurrence ($a = T = b = 1$; phase $\hat{\tau} = 0.5$; 500 realisations at $c_v = 0.30$). Shaded bands show the 10th–90th percentile range; coloured lines are ensemble medians; the black dashed line is the deterministic CSS; vertical dashed lines mark $\hat{x} = \pm 1$. **(a)** Only T_j randomised. **(b)** Only b_j randomised. **(c)** Both T_j and b_j randomised independently. **(d)** Plate-rate-conserving coupling $b_j = v_{\text{plate}} T_j$. In all four cases the ensemble median coincides with the deterministic CSS; see text for the physical interpretation of the band shapes.

locity b/T, and the cycle phase $\hat{\tau} = \tau/T$), a geodetic dataset that samples the near-rift gradient and the far-field plateau simultaneously constrains all three independently, with no locking depth or prescribed far-field forcing required. The gradient position fixes aT; the long-distance plateau fixes b/T; and their combination yields a and T individually, from which $\eta = EH/[2a(1-\nu^2)]$ follows once E and H are estimated from seismological constraints. The stochastic analysis of Section 4.4 quantifies how cycle-to-cycle scatter in T_j and b_j propagates into uncertainty on these inferred parameters, providing a principled uncertainty budget for geodetically constrained viscosity estimates at any magmatic rift system.

Real rifts are segmented rather than homogeneous: dike intrusions span finite along-strike lengths, individual segments differ in recurrence history and plate thickness, and lateral variations in heat flow modulate the effective viscosity along the rift axis. The 1D framework developed here is immediately applicable to segmented geometries because its only length scale is the local product aT, set by the physical properties of each segment. Spatial variations in H or η along strike modulate aT locally without introducing new boundary conditions or requiring a finite model domain: each segment generates its own deformation halo, and the surface velocity field is their superposition. The stochastic ensemble of Section 4.4 already provides the natural language for this multi-segment setting: adjacent segments with different mean recurrence intervals T and variable event sizes b_j contribute independently to the observed velocity profile, and the CSS remains an unbiased predictor of the long-term average field as long as $\mathbb{E}[T_j] = T$ and $\mathbb{E}[b_j] = b$ hold locally for each segment. Extending the CSS to the full 2D problem (accounting for along-strike propagation of dike intrusions and rift segment interaction) is a natural next step, requiring the relaxation of the plane-strain approximation while retaining the same non-local Stokes coupling at the base of the elastic plate.

5 CONCLUSIONS

The CSS framework resolves the kinematic gap between post-diking transients and secular plate motion by recasting inter-rifting deformation as the time-averaged superposition of visco-elastic step responses to a long history of discrete dike intrusions. Because the model satisfies far-field plate divergence by construction and requires no locking depth, prescribed domain width, or external velocity offset, it provides a mechanically self-consistent alternative to backslip and dislocation models across any magmatic rift system.

At Krafla, the viscous CSS fits the post-1984 differenced displacement record within measurement noise ($\chi^2_r = 0.72$ at $T = 200$ yr), yielding $a = 5.17 \times 10^{-5}$ m s^{-1} and an implied asthenospheric viscosity $\eta \approx 4.1 \times 10^{18}$ Pa s ($H = 10$ km, $E = 40$ GPa), consistent with post-glacial rebound estimates beneath Iceland and reflecting the anomalously hot sub-lithospheric mantle maintained by the Iceland plume. Its Maxwell counterpart, with τ_R derived from the same shared-modulus closure (46) rather than fit independently, is only marginally disfavoured (ΔAIC $\approx$ 1.1–1.2): the 8–11 yr Krafla record does not require a Maxwell asthenosphere. At Asal, the viscous CSS again fits well ($\chi^2_r \approx 1.54$) and de-

livers $a = 9.47 \times 10^{-6}$ m s^{-1} and $\eta \approx 2.3 \times 10^{19}$ Pa s, a factor of five higher than at Krafla, in line with the lower heat flux of the Afar setting; here the comparison is decisive (ΔAIC $\approx$ 28–33 in favour of the viscous CSS), because the 25-yr record samples enough of the post-diking transient to rule out a spurious elastic contribution from the derived τ_R.

The non-local Stokes kernel underlying the viscous CSS produces Lorentzian velocity tails that decay more slowly with distance than the erfc profile of Foulger-type diffusion. This structural difference is directly observable in geodetic data: the viscous CSS outperforms the diffusion model in reproducing velocities at stations several kilometres from the rift axis, where the erfc profile underestimates the signal ($\chi^2_r \approx$ 15.7–15.9 versus 33 at Asal). The Asal velocity transect provides the clearest discrimination between the two formulations; at Krafla, the differenced-displacement observable $\Delta u = u(\tau_f) - u(\tau_i)$ suppresses far-field signal regardless of the spatial functional form, so a longer or denser geodetic transect would be required there to separate the two models statistically.

Stochastic ensembles demonstrate that the CSS is an unbiased estimator of the long-term inter-rifting velocity field even at $c_v = 0.30$ cycle-to-cycle variability. Scatter in recurrence interval shifts the velocity gradient laterally, concentrating uncertainty in the transition zone; scatter in dike opening rescales the profile amplitude uniformly, leaving the gradient position unaffected. Coupling the two sources through the plate-rate constraint $b_j = v_{\rm plate} T_j$ eliminates far-field uncertainty entirely. In all cases the ensemble median coincides with the deterministic CSS, confirming that stochastic recurrence introduces spread but not bias.

These results establish the CSS as a practical tool for constraining sub-lithospheric viscosity from geodetic inter-rifting velocity profiles without auxiliary rheological assumptions. Jointly inverting the gradient position and far-field amplitude of a single velocity profile yields the deformation zone half-width aT, the plate velocity b/T, and the ratio EH/η simultaneously. Extending the framework to segmented rifts and to the full along-strike 2D geometry is the direct path toward a system-scale model of strain accumulation at magmatic plate boundaries, retaining the mechanically self-consistent structure derived here.

APPENDIX A: WHY THE STEP RESPONSE SETTLES AT $B/2$, AND WHY B REMAINS THE PHYSICAL DIKE OPENING

Equation (17) predicts that $u^{\rm STEP}(x,t) \to b/2$ as $t \to \infty$ at any fixed x, half the opening imposed by the initial condition (11). This appendix shows that the missing half is not lost: it is a direct, necessary consequence of the antisymmetry of the governing equation, and the opening measured at the intrusion site itself remains exactly b for as long as the model is evaluated.

A1 The initial condition splits into a stationary and an evolving part

The Heaviside initial condition (11) decomposes uniquely into a constant background and an odd perturbation about it:

$$u(x,0^+) = \underbrace{\frac{b}{2}}_{\text{even}} + \underbrace{\frac{b}{2}\,\mathrm{sgn}(-x)}_{\text{odd}}. \tag{A.1}$$

The governing equation (4) constrains u only through its spatial derivatives – the elastic stress (1) depends on $\partial u/\partial x$, and the Stokes traction (3) on $\partial u/\partial t$ integrated against a kernel that itself vanishes for a spatially uniform field. A constant offset therefore carries no stress and satisfies (4) identically at all times: the even term in (A.1) is an exact stationary solution, not an approximation. Equivalently, in Fourier space (10), the even term is carried entirely by the $\delta(k)$ singularity in $\hat{u}(k,0)$, which sits at $k = 0$ where the decay rate $a|k|$ in (10) vanishes – this is exactly the term $I_1 = b/2$ isolated in §2.3. Only the odd term, carried by the regular part ib/k of the spectrum, decays through (10) and gives rise to the arctangent profile (17). The half that appears to vanish in the $t \to \infty$ limit was never dynamical: it is a rigid relabelling of the origin that the elastic–viscous system cannot act on, because absolute displacement carries no physical meaning in an equation that depends only on gradients of u.

A2 The opening at the source is conserved exactly

The physically meaningful, gauge-independent quantity is not the value of u at a fixed point but the discontinuity of u across the intrusion site. Taking $x \to 0^+$ and $x \to 0^-$ in (17) at any fixed $t > 0$, the argument at/x diverges to $\pm\infty$ regardless of how small t is, so

$$\lim_{x\to 0^+} u^{\rm STEP}(x,t) = \frac{b}{2}, \qquad \lim_{x\to 0^-} u^{\rm STEP}(x,t) = -\frac{b}{2}, \qquad \forall\, t > 0, \tag{A.2}$$

and therefore

$$\lim_{x\to 0^+} u^{\rm STEP}(x,t) - \lim_{x\to 0^-} u^{\rm STEP}(x,t) = b, \qquad \forall\, t > 0. \tag{A.3}$$

The opening at the source is not b only in the instant after intrusion; it is b identically, for every $t > 0$, because the odd term in (A.1) is itself odd about $x = 0$ at every later time – relaxation redistributes it in x but cannot break its antisymmetry. Nothing of the imposed slip is dissipated; the model simply reports, at $x = 0$, the two one-sided limits of a field that is continuous everywhere except at the source.

A3 The far-field $b/2$ is half of the opening having diffused outward, not the whole of it

Equation (10) treats $+k$ and $-k$ identically, so the Stokes kernel (3) has no preferred direction: as the odd term in (A.1) relaxes outward from the source, it supplies equal amounts of strain to the $x > 0$ and $x < 0$ half-spaces at every instant. A given point x only registers the full opening b while it remains close to the source relative to the relaxed length scale at; once $at \gg x$, that point lies deep inside the relaxed region and has been supplied with exactly half of the total redistributed opening, the other half having gone to the mirror point $-x$. This is precisely what the self-similar dependence of (17) on $x/(at)$ encodes: for any

fixed x, that ratio vanishes as $t \to \infty$, so every station eventually reports the same $\pm b/2$ plateau regardless of how far it sits from the rift axis. The $b/2$ asymptote is therefore a statement about how the opening is shared between the two flanks in the fully-relaxed limit, not about how much of it survives.

This structure is not particular to the present closure. The same $\pm$(slip)$/2$ far-field split appears in the classical screw-dislocation model of strain accumulation across a locked fault (Savage 1983): a fault offset s produces displacements of $+s/2$ and $-s/2$ on either side, and the full offset s is recovered only as the difference between the two flanks, never as the amplitude at either one alone. Equation (17) inherits the same structure because §2.2 reduces the elastic–viscous coupling to the same non-local, parity-symmetric kernel that generates the dislocation solution.

A4 Consequence for the fitted parameter b

Because the opening at the source is conserved exactly (A.3), the parameter b recovered by fitting (17), its cyclic sum (19), or the Maxwell and CSS forms built on the same kernel (§2.5, §2.4) to geodetic data is the dike-opening width itself, directly comparable to field-mapped intrusion widths without rescaling. This holds specifically for b as a model parameter, not for the displacement recorded at any single station: a station far from the rift axis will, in the long-time limit, register only $\pm b/2$ of the opening, and comparing that single-station amplitude against a field-mapped dike width would underestimate it by up to a factor of two. The two quantities – the fitted b and a far-field station's own displacement – coincide only within the near-field region where $at \lesssim |x|$.

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Data Availability

The geodetic data used in this study are from published sources cited in the text. The analytical model code and inversion scripts are available at [PLACEHOLDER — repository DOI or URL upon acceptance].

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